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The Incident Detection Message Exchange Format version 2 (IDMEFv2)

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Abstract

The Incident Detection Message Exchange Format version 2 (IDMEFv2) defines a data representation for security incidents detected on cyber and/or physical infrastructures.

The format is agnostic so it can be used in standalone or combined cyber (SIEM), physical (PSIM) and availability (NMS) monitoring systems. IDMEFv2 can also be used to represent man made or natural hazards threats.

IDMEFv2 improves situational awareness by facilitating correlation of multiple types of events using the same base format thus enabling efficient detection of complex and combined cyber and physical attacks and incidents.

This draft is maintained by the IDMEFv2 Task Force. Please consult our website for more information: https://www.idmefv2.org.

If approved this draft will obsolete RFC4765.

Status of this Memo

This Internet-Draft is submitted in full conformance with the provisions of BCP 78 and BCP 79.

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1. Introduction

The Incident Detection Message Exchange Format (IDMEF) is intended to solve the problem of security monitoring compartmentalization by proposing a single format to represent any type of incident, whether cyber or physical, intentional or accidental, natural or man-made.

Indeed security is often associated to the Confidentiality-Integrity-Availability triad, performance and availability management systems are still run independently from security management systems.

Additionally, with the adoption and integration of Internet of Things (IoT) and Industrial Internet of Things (IIoT) devices, and the exponential emergence of smart systems (transport, cities, buildings, etc), an increasingly interconnected mesh of cyber-physical systems (CPS) has emerged. This expansion of the attack and incident surfaces blurs the once-clear functions of cybersecurity and physical security.

Finally, as IT infrastructure moves out of data centers it becomes more exposed to external threats, including natural and man-made hazards.

Incident detection systems have traditionally focused on detecting cyber incidents or physical incident or availability incidents. There is an increasing need nowadays to have a unified view and management of all those incidents and their interconnection.

To achieve this goal the Incident Detection Message Exchange Format offers a unique data representation for multiple types of events:

- Cyber-security events (e.g. authentication failure/success, virus/malware detection, bruteforce/scan detection, etc.)
- Physical security events (e.g. intrusion detection, object detection, face or activity recognition, fire/smoke/noise/rain detection, etc.)
- Availability/observability/performance events (e.g. system failure, service malfunction, performance decrease, etc.)
- Natural and man made hazards events (e.g. wildfires, avalanches, droughts, earthquakes, pollution, fire, explosion, etc.)

1.1. IDMEFv2 deployment architecture

IDMEFv2 can be used to exchange incident detection information between specialized managers (SIEM, PSIM, NMS) and a universal "Cyber & Physical SIEM" (CPSIEM) or directly from specialized analyzers and a CPSIEM.

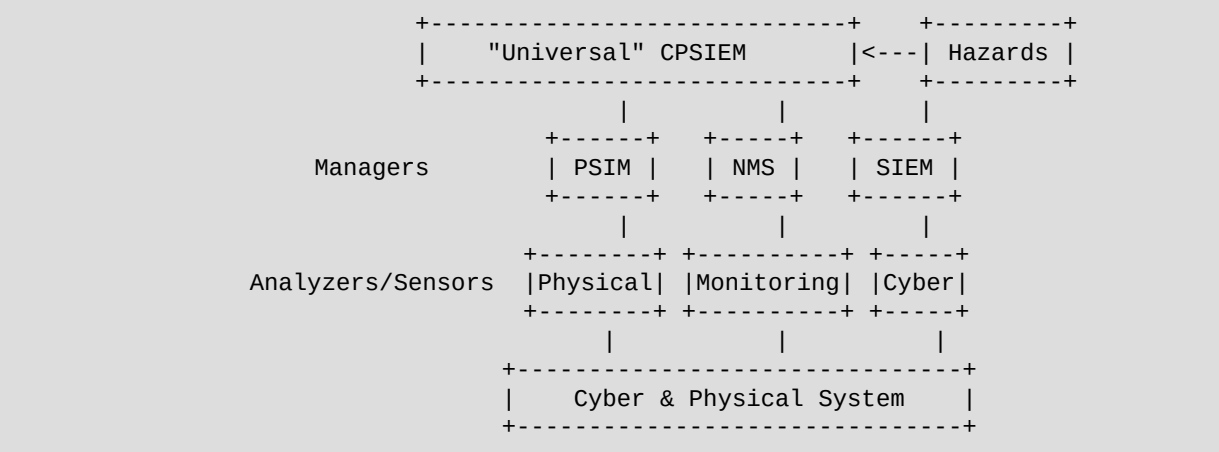


Figure 1: IDMEF Use Architecture

IDMEFv2 improves situational awareness by enabling correlation of multiple types of events using the same base format.

This document defines a model serialization methods for the purpose of describing and sharing these events.

1.2. IDMEFv1 (Intrusion Detection Message Exchange Format) - RFC 4765 - Legacy

IDMEFv2 (Incident Detection Message Exchange Format) is based on IDMEFv1 (Intrusion Detection Message Exchange Format) concepts. But IDMEFv1 was cyber intrusion focused as IDMEFv2 perimeter is much larger. Thus retro-compatibility although partly possible has not been a priority.

1.3. Relationship between IDMEFv2 and other event/incident formats

IDMEFv2 focuses essentialy on high level event/incident correlation and detection.There are many standard and proprietary formats on the incident detection market and in particular on the cybersecurity market. IDMEFv2 is complementary to most of these formats.

IDMEFv1 (Intrusion Detection Message Exchange Format - RFC 4765) : IDMEFv2 (Incident Detection) replaces and obsoletes IDMEFv1 (Intrusion Detection) by covering a wider spectrum.

IODEFv2 (Incident Object Definition Exchange Format - RFC 5070) : IDMEFv2 helps detect incident. When an incident is detected it will be analysed and eventually fully described and shared with other security teams through IODEFv2. IDMEF is used upstream IODEFv2. IDMEFv2 alerts can be “attached” to IODEFv2 object to provide technical details about incidents.

Syslog (System Logging) : Syslog is a lossy format with no formal structure. Syslog can be used by sensors to send information to analyzers. Out of those multi-format syslogs the analyzer might detect an incident or an event of interest. The analyzer will then use IDMEFv2 to notify the manager which might correlate this information with other data to confirm the incident.

SNMP (Simple Network Management Protocol) : SNMP polls information from devices which is then compared to thresholds to detect incident. IDMEFv2 can be used when incident is detected downstream of SNMP to communicate the incident to the manager. IDMEFv2 can have a similar role as SNMP Traps.

STIX (Structured Threat Information Expression) : is a language and serialization format used to exchange cyber threat intelligence (CTI). IDMEFv2 can help detect incidents which might lead to the creation and sharing of STIX information. Cyber analyzer can also rely on STIX information to detect incidents that will be notified in IDMEFv2 format.

OCSF (The Open Cybersecurity Schema Framework) is an open-source, vendor-agnostic standard designed to normalize security telemetry from diverse tools. It provides a common language and consistent structure for security event data, simplifying data ingestion, correlation, and analysis. OCSF can be seen as a “super” syslog to describe events before IDMEFv2 extract “incidents”. OCSF is limited to cyber security.

SIEM proprietory formats (CEF, LEEF, ECS, CIM, ...) : By covering cyber, physical and monitoring incidents type, IDMEFv2 offers a wider spectrum than those formats. Gateways between IDMEFv2 and those formats can be developped to connect legacy cyber detection systems to an IDMEFv2 architecture.

1.4. Existing Deployments and Adoption

IDMEFv2 is not a theoretical proposal. It has been developed, validated, and deployed within the framework of eight large-scale European research projects, funded by the Horizon 2020 and Digital Europe programmes. These projects — namely 7SHIELD, PRECINCT, CyberSEAS, ATLANTIS, ENDURANCE, KINAITICS, TESTUDO, and SAFE4SOC — address critical sectors such as space systems, energy grids, transportation, and government infrastructure. These implementations span multiple domains, including Security Operations Centers (SOCs), Physical Security Information Management (PSIM) systems, and critical infrastructure protection pilots.

This document aims to formalize this existing practice as an IETF Experimental RFC, to ensure interoperability, gather broader community feedback, and provide a stable foundation for future developments.

2. Terminology

2.1. Keywords

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [RFC2119].

2.2. Normative sections

Implementations of IDMEFv2 are REQUIRED to fully implement:

- The data types defined in Section 3
- The data model defined in Section 5
- The JavaScript Object Notation (JSON) serialization method Section 5.8.

2.3. Concepts related to event processing

2.3.1. Event

An event is something that triggered a notice. Any incident starts off as an event or a combination of events, but not all events result in an incident. An event does not need be an indication of wrongdoing. E.g. someone successfully logging in or entering a building is an event.

2.3.2. Incident

An incident is an event that compromises or has a significant probability of compromising at least one of the organization's security criteria such as Confidentiality, Integrity or Availability. An incident may affect a production tool, personnel, etc. It may be logical, physical or organizational in nature. Last but not least, an incident may be caused on purpose or by accident.

2.3.3. Alert

An alert is a notification/message that a particular event/incident (or series of events/incidents) has occurred.

2.3.4. Manager

The manager is the central console toward which all analyzers send their alerts. The manager collects, correlates, stores and displays the alerts to the operators.

Example : - A SIEM (Security Information & Event Management) or a Log Manager) - A PSIM (Physical Security Information Management) - A NMS (Network Management System) - A CPSIEM (Cyber & Physical Security Information Management System)

2.3.5. Operator

The level 1 operator is in charge of receiving manager notifications and identify or confirm when an event should be considered as an incident. The operator must also decide if there is a known resolution for this incident or if it needs a deeper analysys.

2.3.6. Analyst

The analyst will be contacted by the operator to analyze complex incidents that can’t be easily resolved. The investigation starts with the IDMEFv2 information but the analyst might need more information like raw logs for a deeper forensics.

2.3.7. Attack

An attack is an attempt to destroy, expose, alter, disable, steal or gain unauthorized access to or make unauthorized use of a cyber or physical asset. An attack is one or many kinds of incidents.

2.3.8. Correlation

Correlation is the identification of relationships between two or more events.

2.3.9. Aggregation

Aggregation is the consolidation of similar events into a single event.

3. The IDMEF Data Types

Each object inside the IDMEF data model has an associated data type. This type may be used to validate the content of incoming IDMEF messages.

3.1. Classes

The classes are meant to group related attributes together. Some of the classes may be instanciated multiple times (e.g. Source, Target, etc.) while others may only appear once in an IDMEF message (e.g. Analyzer).

3.2. Numbers

3.2.1. Integers

Integers inside the IDMEF data model are expressed using the following ABNF [RFC5234] grammar:

```
integer         =  *1minus int
int             =  zero / ( digit1-9 *DIGIT )
minus           =  %x2D                          ; -
zero            =  %0x30                         ; 0
digit1-9        =  %x31-39                       ; 1-9
```

E.g. 123.

Such values are indicated with the "INT" type annotation in the model.

3.2.2. Floating-point values

Floating-point values inside the IDMEF data model are expressed using the following ABNF grammar:

```
float           =  integer *1frac
frac            =  decimal-point 1*DIGIT
decimal-point   =  %x2E                           ; .
```

This grammar reuses some of the production rules listed in Section 3.2.1.

E.g. 12.34.

Such values are indicated with the "FLOAT" type annotation in the model.

3.3. Strings

Strings are series of characters from the [UNICODE] standard and are used to represent a text.

For readability, this document uses quotes (") to delimit strings, but please note that these quotes are not syntactically part of the actual strings.

E.g. "Hello world".

Some of the strings used in the IDMEFv2 data model follow a stricter syntax. These are included below for completeness.

Such values are indicated with the "STRING" type annotation in the model.

3.3.1. Enumerations

Enumerations are special strings used when valid values for an IDMEF attribute are restricted to those present in a predefined list.

Such values are indicated with the "ENUM" type annotation in the model.

3.3.2. Timestamps

Timestamps are used to indicate a specific moment in time. The timestamps used in the IDMEF data model follow the syntax defined by the "date-time" production rule of the grammar in [RFC3339] ch 5.6.

E.g. "1985-04-12T23:59:59.52Z" represents a moment just before April 5th, 1985 in Coordinated Universal Time (UTC).

Such values are indicated with the "TIMESTAMP" type annotation in the model.

3.3.3. Geographical Locations

Some attributes inside the IDMEF data model may refer to geographical locations using a set of coordinates. The reference system for all geographical coordinates is a geographic coordinate reference system, using the World Geodetic System 1984 [WGS84]. The reference system used is the same as for the Global Positioning System (GPS).

The format for such values can be either "latitude,longitude" or "latitude,longitude,altitude". Each of these coordinates is represented as a floating-point value. The latitude and longitude are expressed in degrees while the altitude is expressed in meters.

E.g. "48.8584,2.2945,276.13" matches the (3-dimensional) geographical location for the top floor or the Eiffel Tower located in Paris, France, while "48.8584,2.2945" matches the same location in two dimensions (with the altitude removed).

Such values are indicated with the "GEOLOC" type annotation in the model.

3.3.4. UNECE Location Codes (UN/LOCODE)

Some attributes inside the IDMEF data model may refer to geographical locations using Locations Codes. These codes can be assimilated to an enumeration, where the list of possible values is defined in the United Nations Economic Commission for Europe (UNECE) Codes for Trade [UN-LOCODE].

E.g. "FR PAR" is the Location Code for the city of Paris, France.

Such values are indicated with the "UNLOCODE" type annotation in the model.

3.3.5. Uniform Resource Identifiers (URIs)

The IDMEF data model uses Uniform Resource Identifiers (URIs), as defined in [RFC3986], when referring to external resources. Unless otherwise specified, either a Uniform Resource Location (URL) or a Uniform Resource Name (URN) may be used where a URI is expected.

E.g. both "https://example.com/resource" and "urn:myapp:resource" are valid Uniform Resource Identifiers.

Such values are indicated with the "URI" type annotation in the model.

3.3.6. IP Addresses

IP addresses inside the IDMEF data model are expressed as strings using the traditional dotted-decimal notation for IPv4 addresses (defined by the "dotnum" production rule in the grammar in [RFC5321]), while IPv6 addresses are expressed using the text representation defined in [RFC4291] ch 2.2.

E.g. "192.0.2.1" represents a valid IPv4 address, while "::1/128" represents a valid IPv6 address.

It is RECOMMENDED that implementations follow the recommendations for IPv6 text representation stated in [RFC5952].

Such values are indicated with the "IP" type annotation in the model.

3.3.7. E-mail addresses

E-mail addresses inside the IDMEF data model are expressed as strings using the address specification syntax defined in [RFC5322] ch 3.4.1.

E.g. "root@example.com".

Such values are indicated with the "EMAIL" type annotation in the model.

3.3.8. Attachment names

Attachments inside the IDMEF data model are identified using a unique name, composed of a string whose character set is limited to the ASCII letters (A-Z a-z) and digits (0-9).

E.g. "state" is a valid name for an attachment.

The constraint on name unicity is enforced per class. That is, but it is not possible for two attachments to share the same name inside the same alert.

Such values are indicated with the "ID" type annotation in the model.

3.3.9. Media types

Media types are used in the IDMEF data model to describe an attachment's content. The syntax for such values is defined in [RFC2046].

IANA keeps a list of all currently registered media types in the Media Types registry .

E.g. "application/xml" or "text/plain; charset=utf-8".

Such values are indicated with the "MEDIATYPE" type annotation in the model.

3.3.10. Universally Unique IDentifiers (UUIDs)

Universally Unique Identifiers (UUIDs) are used to uniquely identify IDMEF messages. It is also possible for an IDMEF message to reference other IDMEF messages using their UUIDs. The syntax for UUIDs is defined in [RFC4122].

To limit the risk of UUID collisions, implementors SHOULD NOT generate version 4 UUIDs (randomly or pseudo-randomly generated UUIDs).

E.g. "ba2e4ef4-8719-42bb-a712-d6e8871c5c5a".

UUIDs are case-insensitive when used in comparisons.

Such values are indicated with the "UUID" type annotation in the model.

3.3.11. Protocol Names

Such values are indicated with the "PROTOCOL" type annotation in the model.

3.3.12. IDMEF Paths

This document defines a way to represent the path to every possible attribute inside an IDMEF message. For conciseness, the top-level "Alert" class is omitted from the path.

This representation can be used in contexts where the path to an IDMEF attribute is expected. An example of such usage can be seen in the definition of the "AggrCondition" attribute inside the Alert class (Section 5.2).

The syntax for these IDMEF paths is expressed in the following ABNF grammar:

```
class-name      =  "Analyzer" / "Sensor" / "Source" / "Target" /
                   "Attachment"
attribute-name  =  1*ALPHA
class-reference =  class-name "."
num             =  *1"-" 1*DIGIT
list-index      =  "(" num ")"
path            =  *1class-reference attribute-name *1list-index
```

Valid attribute names are limited to those defined for the specified class-reference (or in the top-level "Alert" class if class-reference is omitted).

For example, the following path refers to the "CeaseTime" attribute of the top-level "Alert" class: "CeaseTime".

Likewise, the following path refers to the "Name" attribute of the "Analyzer" class: "Analyzer.Name".

For attributes defined as lists (see Section 3.4), the path may include the (0-based) index for an entry inside the list. The index defaults to 0 if omitted. This means that several (valid) representations may be used to reference the same IDMEF attribute when list attributes are involved.

For example, both of the following paths refer to the IP address of the first source associated with an IDMEF message:

```
Source.IP
Source(0).IP
```

Compatible implementations MUST reject paths that reference an unknown class, an unknown attribute, or use a list-index for an IDMEF field which is not defined as a list.

A compatible implementation MUST also normalize paths before comparing them (e.g. by stripping the text "(0)" from paths referring to list attributes).

3.3.13. Hashes

Hashes are sometimes used inside the data model to protect the integrity (and optionally, authenticity) of attachments.

The syntax for these values is "function:hash_result", where "function" refers to one of the hashing function names listed in and "hash_result" contains the hexadecimal notation for the hash result obtained by calling the specified hash function on the input value.

In the context of IDMEF, either a keyless or keyed hash function may be used to process the raw input value.

E.g. "sha256:a02735ed8b10ad432d557bd4849c0dac3b23d64706e0618716d6df2def338374"

Hashes are case-insensitive when used in comparisons.

Such values are indicated with the "HASH" type annotation in the model.

3.4. Lists

Some attributes of the IDMEF data model accept ordered lists of values.

Such ordered lists are indicated with the "X[]" type annotation in the model. where "X" refers to one of the data types defined in Section 3. For example, "ENUM[]" refers to an ordered list of enumeration values.

4. The IDMEF extension

In order to support the dynamic nature of security operations and to adapt to specific needs, the IDMEFv2 data model will need to continue to evolve. This section discusses how new data elements can be incorporated into the IDMEFv2. There is support to add additional enumerated values and new attributes.

These extension mechanisms are designed so that adding new data elements is possible without requiring modifications to this document. Extensions can be implemented publicly or privately. With proven value, well-documented extensions can be incorporated into future versions of the specification.

4.1. Extending the Enumerated Values of Attributes

Additional enumerated values can be added to select attributes either through the use of specially marked attributes with the "ext-" prefix or through a set of corresponding IANA registries. The former approach allows for the extension to remain private. The latter approach is public.

4.1.1. Private Extension of Enumerated Values

The data model supports adding new enumerated values to an attribute without public registration. For each attribute that supports this extension technique, there is a corresponding attribute in the same element whose name is identical but with a prefix of "ext-". This special attribute is referred to as the extension attribute. The attribute being extended is referred to as an extensible attribute. For example, an extensible attribute named "foo" will have a corresponding extension attribute named "ext-foo". An element may have many extensible attributes.

In addition to a corresponding extension attribute, each extensible attribute has "ext-value" as one its possible enumerated values. Selection of this particular value in an extensible attribute signals that the extension attribute contains data. Otherwise, this "ext-value" value has no meaning.

In order to add a new enumerated value to an extensible attribute, the value of this attribute MUST be set to "ext-value", and the new desired value MUST be set in the corresponding extension attribute. For example, extending the Category attribute of the Analyzer class would look as follows:

```
Analyzer: {
    ...
    "Category":["ext-value"],
    "ext-Category": "my-new-analyzer-category",
    ....
}
```

A given extension attribute MUST NOT be set unless the corresponding extensible attribute has been set to "ext-value".

4.1.2. Public Extension of Enumerated Values

The data model also supports publicly extending select enumerated attributes. A new entry can be added by registering a new entry in the appropriate IANA registry. Section (Table 5) provides a mapping between the extensible attributes and their corresponding registry.

4.2. Private Extension of Attributes

Use of new attributes is possible through the use of the attachment class. New attributes and their corresponding values should be stored in the Content attribute of an Attachment and the ContentEncoding must be set to JSON. For example creating a new attribute to store the email of the operator (in charge of solving the incident) will look as follows:

```
"Attachment" : [
    {
    "Name": "Operator",
    "ContentEncoding": "JSON",
    "Content": "{\"OperatorMail\":\"John.Does@acme.com\"}",
    }
[
```

5. The IDMEF Data Model

In this section, the individual components of the IDMEF data model will be discussed in detail. For each class, the semantics will be described.

5.1. Overview

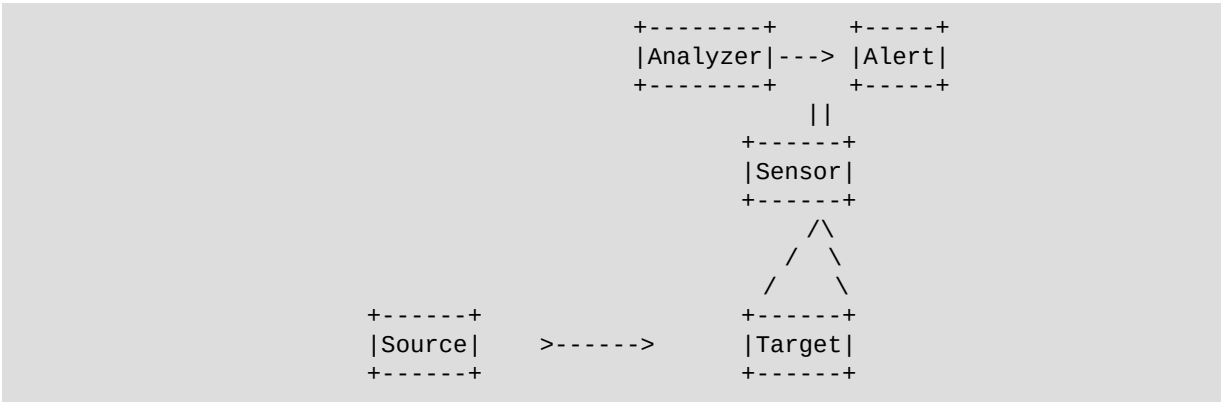


Figure 2: IDMEFv2 Overview Classes

An IDMEF message is composed of an instance of the Alert class (Section 5.2) representing the overall properties of the message. It also contains exactly one instance of the Analyzer class (Section 5.3) and zero or more instances of the Sensor class (Section 5.4). The message may also describe various aspects of an event using the Source (Section 5.5) and Target (Section 5.6) classes.

Last but not least, it may also include zero or more instances of the Attachment class (Section 5.7), e.g. captured files or network packets related to the event for example.

The relationship between the main Alert class and other classes of the data model is shown in Figure 3 (attributes are omitted for clarity).

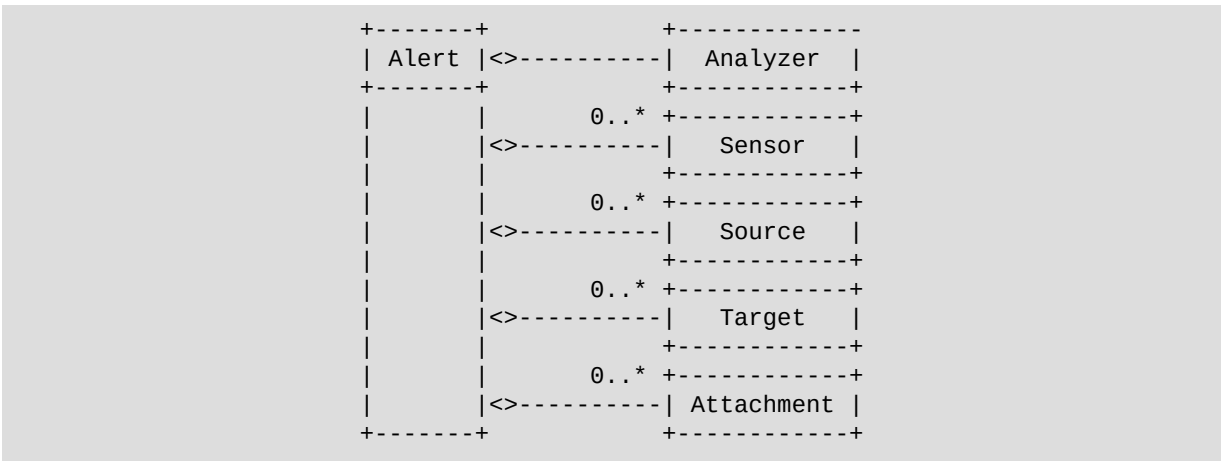


Figure 3: IDMEFv2 Classes

It is important to note that the data model does not specify how an incident should be categorized or identified. For example, an attacker scanning a network for machines listening on a specific port may be identified by one analyzer as a single attack against multiple targets, while another analyzer may identify it as multiple attacks from a single source. However, once an analyzer has determined the type of alert it plans on sending, the data model dictates how that alert should be formatted.

5.2. The Alert Class

The Alert class contains high level information about the event that triggered the alert.



Figure 4: The Alert class

The aggregate classes that make up Alert are:

Analyzer
: Exactly one. An instance of the Analyzer class (Section 5.3) that describes the tool/device responsible for the analysis that resulted in the alert being created and sent.

Sensor
: Zero or more. Instances of the Sensor class (Section 5.4) used to describe the sensor(s) that captured the information used during the analysis.
Depending on the tools/devices used to detect incidents, an Analyzer may rely on the output from a single sensor or from multiple sensors to generate alerts. In addition, the Analyzer and Sensor may actually be part of the same physical device and may share some of their attributes (e.g. IP, Hostname, Model, etc.).

Source
: Zero or more. Instances of the Source class (Section 5.5) used to describe the source(s) of the incident (e.g. attackers, faulty device, etc.).

Target
: Zero or more. Instances of the Target class (Section 5.6) used to describe the target(s) of the incident, i.e. the impacted devices/users/services/locations.

Attachment	Zero or more. Instances of the Attachment class (Section 5.7) used to describe the electronic artifacts captured in relation with the event. The intent of the Attachment class is to keep track of the electronic files left as a trail during the event. This may include things like on-disk files (e.g. malware samples), network packet captures, videos or still images from a camera feed, voice recording, etc.

The Alert class has the following attributes:

Version	Mandatory. The version of the IDMEF format in use by this alert. During the drafts tuning period the version is equal to the draft version. Therefore it is "2.D.V0X" for Draft V0X.
ID	Mandatory. Unique identifier for the alert.
OrganisationName	Optional. Corporate/Main Office Organisation Name Useful if alerts are sent to a multi-organisation central incident detection manager. Example: ACME Corporation
OrganisationId	Optional. Corporate/Main Office Organisation ID. Where possible official organisation ID manage by national authority. Useful if alerts are sent to a multi-organisation central incident detection manager. This ID has to be chosen depending on the overall detection perimeter and the nature of the monitored organisation (Private/Public, Commercial, International, etc.) Examples: OrganisationId in France could be SIREN, in England could be CR, Germany could be Handelsregisternummer, Spain could be CIF, Italia could be Partita IVA, USA could be EIN, etc. Commercial OrganisationId in Europe could be V.A.T ID
EntityName	Optional. Entity Name, monitored by the organisation, where the incident occurred. Could be a town, region or country name or an internal name. Could also be the name of a customer for a MSSP centralizing it's customer incidents in a single system. Do not repeat the organisation name in the EntityName Example: - ACME HeadQuaters is located in Paris France and has a local office in India - If the incident occurred in the local office: "OrganisationName": "ACME" , "EntityName": "India" - If the incident occurred in the headquaters: "OrganisationName": "ACME", "EntityName": "Headquaters" (or "Paris")
EntityId	Optional. Entity ID, monitored by the organisation, where the incident occurred. Useful if organisation and entity are not directly linked, like a customer and a MSSP.
EntitySector	Optional. The economic sector(s) and sub-sector(s) in which the entity operates. Values follow the dot notation sector.subsector based on the critical infrastructure taxonomy defined in the NIS2 Directive and CER (Critical Entities Resilience) Directive. This attribute enables sector-based correlation, regulatory compliance reporting, and risk context for incident detection. Enumeration available here Table 9

Type — Optional. Incident type.

Rank	Keyword	Description
0	Cyber	Incident involving cyberspace, information systems, or digital assets.
1	Physical	Incident involving physical assets, facilities, or infrastructure.
2	Availability	Incident affecting the availability or performance of a service, system, asset or resource.
3	Hazards	Incident arising from natural, societal, technological, environmental, or other hazards.
4	Combined	Incident involving two or more incident types.

Table 1: Incident types

Category — Optional. Incident category.
Enumeration available here Table 10

ext-Category — Optional. A means by which to extend the Category attribute. (see Section 4.1.1)

Cause — Optional. Incident cause. The cause can be modified by any analyzer on the way of the alert and later by the operator and/or the analyst if new investigation reveals and confirms a different cause of the event.

Rank	Keyword	Description
0	Normal	The event is related to an expected phenomenon or to a phenomenon that does not qualify as out of the ordinary.
1	Error	The event is related to a human error.
2	Malicious	The event is related to malicious code or malicious actions.
3	Malfunction	The event is related to a device or service malfunction.
4	Hazard	The event is related to a hazard phenomenon.
5	Unknown	The cause of the event is unknown.

Table 2: Incident causes

Description — Optional. Short free text human-readable description of the event.
The description can add detail to the event category for easiest/faster comprehension by the operator. Example : * Cryptoware WannaCry blocked on pegasus server * Unknown person entering through east doorway

Status — Optional. Event state in the overall event lifecycle.

Rank	Keyword	Description
0	Event	The event is still considered as an harmless event and should not be treated.
1	Incident	The event is considered as an incident and should be taken care of.
2	Open	The incident is confirmed and actively being investigated.
3	Closed	Investigation is finished and the issue is handled.
4	FalsePositive	Investigation is finished, incident was a false positive.
5	Obsolete	Incident is obsolete, usually by another incident.

Table 3: Incident statuses

Priority — Optional. Priority of the alert. Priority is defined by combining impact and urgency. It indicates how fast the incident should be taken care of.

Rank	Keyword	Description
0	Unknown	Priority unknown
1	Info	Informational event with no current impact; no action or follow-up is required.
2	Low	Minor incident with negligible impact; a workaround exists, and resolution can be scheduled.
3	Medium	Moderate incident affecting a limited scope; requires timely attention but is not immediately business-threatening.
4	High	Significant incident with major service degradation or impact on important assets; demands an urgent response.
5	Critical	Severe incident with business-critical systems down or vital interests threatened; requires immediate action and escalation.

Table 4: Incident priorities

Confidence — Optional. A floating-point value between 0 and 1 indicating the analyzer's confidence in its own reliability of this particular detection, where 0 means that the detection is surely incorrect while 1 means there is no doubt about the detection made.

Note — Optional. Free text human-readable additional note, possibly a longer description of the incident if is not already obvious.
The Note attribute can be used to store any additional information. It can be additional information about the event and/or about the incident resolution,

	although the incident resolution information should in principle be stored elsewhere (with a link with the external tool in AltNames)
CreateTime	Mandatory. Timestamp indicating when the alert was created.
StartTime	Optional. Timestamp indicating the deduced start of the event. StartTime can be later than CreateTime in case or Alerts created from forecast information (e.g. Snow Storm expected in two days staring at 10h00)
EndTime	Optional. Timestamp indicating the deduced end of the event.
ReportTime	Optional. Timestamp indicating the reporting time of the event, usually to an external CSIRT or a central SIEM.
AltNames	Optional. Alternative identifiers; strings which help pair the event to internal systems' information (for example ticket IDs inside a request tracking systems).
AltCategory	Optional. Alternate categories from a reference other than IDMEFv2 categories (e.g. MISP, MITRE ATT@CK or another proprietary/internal reference).
Ref	Optional. References to sources of information related to the incident and/or vulnerability, and specific to this incident. This MAY be a URL to additional info, or a URN in a registered or unregistered ad-hoc namespace bearing reasonable information value and uniqueness, such as "urn:cve:CVE-2013-2266".
CorrelID	Optional. Identifiers for the messages which were used as information sources to create this message, in case the message has been created based on correlation/analysis/deduction from other messages.
AggrCondition	Optional. A list of IDMEF fields used to aggregate events. The values for these fields will be the same in all aggregated events. This attribute should mostly be set by intermediary nodes, which detect duplicates, or aggregate events, spanning multiple detection windows, into a longer one. The "StartTime" and "EndTime" attributes are used in conjunction with this attribute to describe the aggregation window.
PredID	Optional. A list containing the identifiers of previous messages which are obsoleted by this message. The obsoleted alerts SHOULD NOT be used anymore. This field can be used to "update" an alert.
RelID	Optional. A list containing the identifiers of other messages related to this message.
SignedAttributes	Optional. List of IDMEFv2 paths of the attributes included in the signature.
Signature	Optional. JWS Compact Serialization string containing the signature.

5.3. The Analyzer Class

The Analyzer class describes the module that has analyzed the data captured by the sensors, identified an event of interest and decided to create an alert.

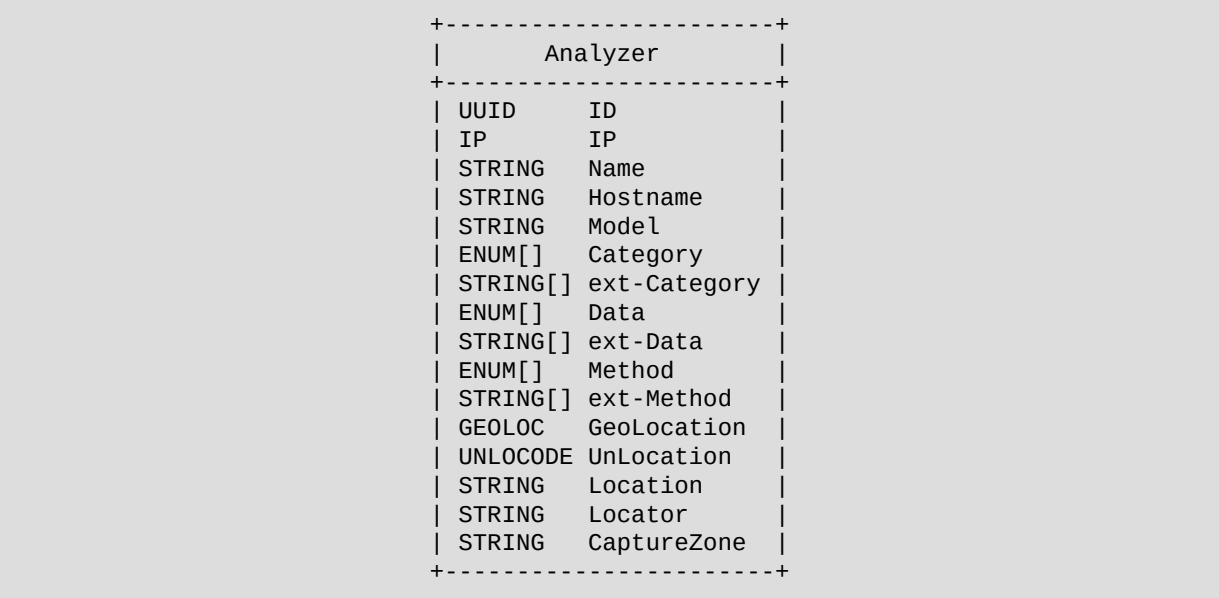


Figure 5: The Analyzer class

The Analyzer class has the following attributes:

ID	Optional. Unique identifier for the analyzer.
IP	Optional. Analyzer IP address.
Name	Recommended. Name of the analyzer, which must be reasonably unique, however still bear some meaningful sense. This attribute usually denotes the hierarchy of organizational units the detector belongs to and its own name. It MAY also be used to distinguish multiple analyzers running with the same IP address.
Hostname	Optional. Hostname of this analyzer. SHOULD be a fully-qualified domain name.
Model	Optional. Analyzer model description (usually its generic name, brand and version).
Category	Optional. Analyzer categories. Enumeration available here Table 11
ext-Category	Optional. A means by which to extend the Category attribute. (see Section 4.1.1)
Data	Optional. Type of data analyzed during the detection. Enumeration available here Table 12
ext-Data	Optional. A means by which to extend the Data attribute. (see Section 4.1.1)
Method	Optional. Detection method. Enumeration available here Table 13
ext-Method	Optional. A means by which to extend the Method attribute. (see Section 4.1.1)
GeoLocation	Optional. GPS coordinates for the analyzer.
UnLocation	Optional. Standard UN/Locode for the analyzer.
Location	Optional. Internal name for the location of the analyzer.
Locator	Optional. Resolvable URL for the entity. May be dynamic or expire; complements Location, which is a persistent path.

CaptureZone	Optional. A string that describes the "capture zone" of the analyzer (if the analyzer is also a sensor), as a JSON-serialized string. Depending on the type of sensor, the capture zone may for instance refer to: A JSON object describing a camera's settings (elevation, horizontal and vertical field of view, azimuth, etc.) A description of the IP network where packet capture is taking place.

5.4. The Sensor Class

The Sensor class describes the module that captured the data before sending it to an analyzer. The Sensor may be a subpart of the Analyzer.

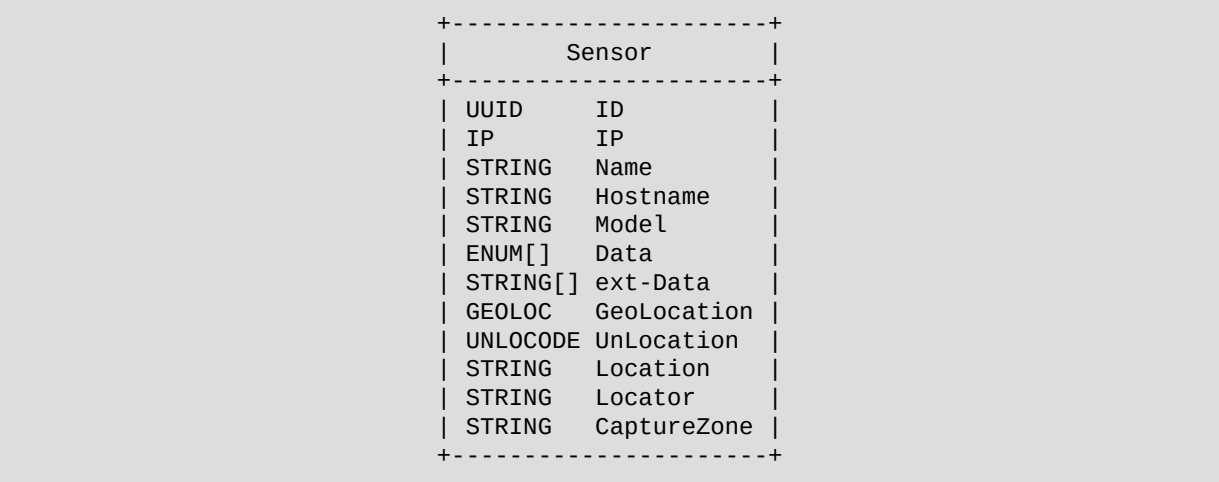


Figure 6: The Sensor class

The Sensor class has the following attributes:

ID	Optional. Unique identifier for the sensor.
IP	Optional. The sensor's IP address.
Name	Recommended. Name of the sensor, which must be reasonably unique, however still bear some meaningful sense. This attribute usually denotes the hierarchy of organizational units the sensor belongs to and its own name. It MAY also be used to distinguish multiple sensors running with the same IP address.
Hostname	Optional. The sensor's hostname. This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed. An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry).
Model	Optional. The sensor model's description (usually its generic name, brand and version).
Data	Optional. Type of data captured by the sensor. Enumeration available here Table 12
ext-Data	Optional. A means by which to extend the Data attribute. (see Section 4.1.1)
GeoLocation	Optional. GPS coordinates for the analyzer.
UnLocation	Optional. Standard UN/Locode for the sensor.
Location	Optional. Internal name for the location of the sensor.

Locator	Optional. Resolvable URL for the entity. May be dynamic or expire; complements Location, which is a persistent path.
CaptureZone	Optional. A string that describes the "capture zone" of the sensor, as a JSON-serialized string. Depending on the type of sensor, the capture zone may for instance refer to: A JSON object describing a camera's settings (elevation, horizontal and vertical field of view, azimuth, etc.) A description of the IP network where packet capture is taking place.

5.5. The Source Class

The Source class describes the origin(s) of the event(s) leading up to the creation of this alert.

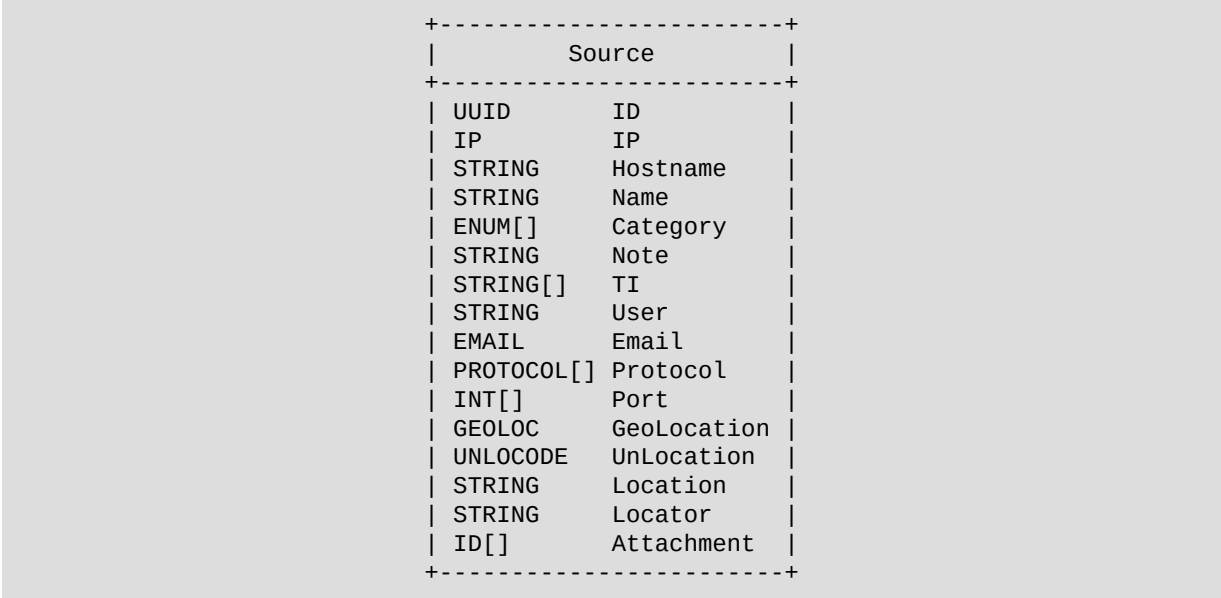


Figure 7: The Source class

The Source class has the following attributes:

ID	Optional. Unique identifier for the source.
IP	Optional. Source IP address.
Hostname	Optional. Hostname of this source. This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed. An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry).
Name	Optional. Name of this source. Naming the source might be necessary for non digital source like a human, a vehicle, a drone, etc.
Category	Optional. Source category Enumeration available here Table 14
Note	Optional. Free text human-readable additional note for this source.

TI	Optional. Threat Intelligence data about the source. Values in this list MUST use the format "attribute:origin", where "attribute" refers to the attribute inside this source found inside a Threat Intelligence database, and "origin" contains a short identifier for the Threat Intelligence database. E.g. "IP:Dshield". Please note that the same attribute may appear multiple times inside the list (because a match was found in multiple Threat Intelligence databases).
User	Optional. User ID or login responsible for the incident.
Email	Optional. Email address responsible for the incident. E.g. the value of the "Reply-To" or "From" header inside a phishing e-mail.
Protocol	Optional. Protocols related to connections from/to this source. If several protocols are stacked, they MUST be ordered from the lowest (the closest to the medium) to the highest (the closest to the application) according to the ISO/OSI model.
Port	Optional. Source ports involved in the incident. Values in this list MUST be integers and MUST be in the range 1-65535.
GeoLocation	Optional. GPS coordinates for the source.
UnLocation	Optional. Standard UN/Locode for the source.
Location	Optional. Internal name for the location of the source.
Locator	Optional. Resolvable URL for the entity. May be dynamic or expire; complements Location, which is a persistent path.
Attachment	Optional. Identifiers for attachments related to this source. Each identifier listed here MUST match the "Name" attribute for one of the attachments described using the Attachment class (Section 5.7).

5.6. The Target Class

The Target class describes the target(s) impacted by the event(s) leading up to the creation of this alert.

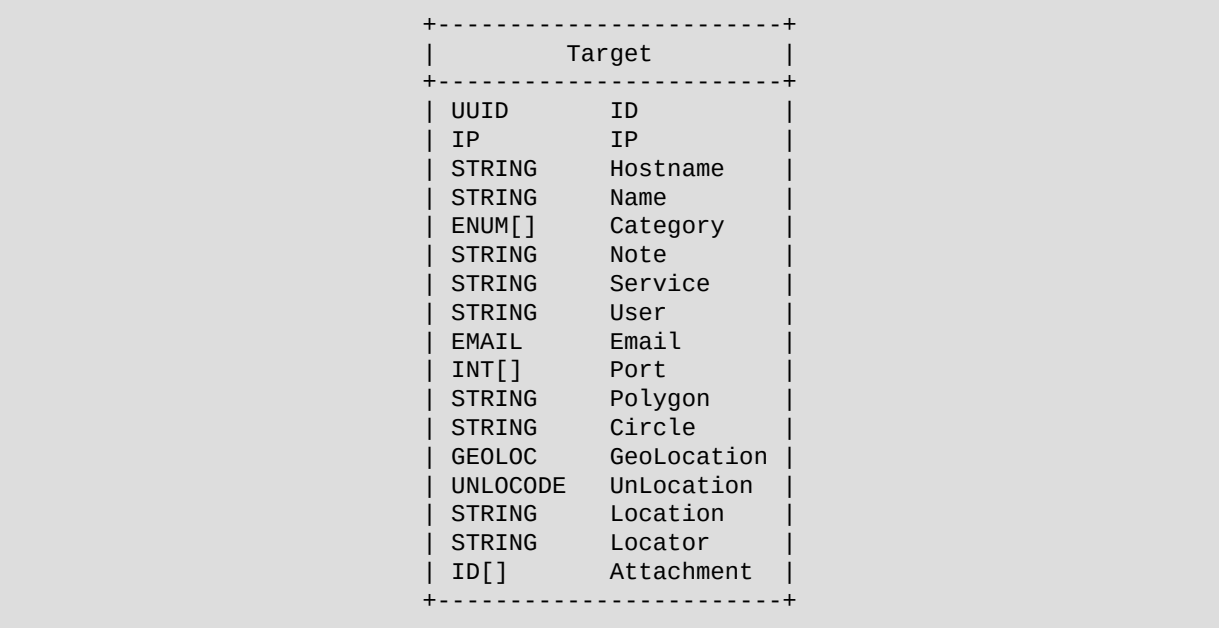


Figure 8: The Target class

The Target class has the following attributes:

ID	Optional.Unique identifier for the target.
IP	Optional. Target IP address.
Hostname	Optional. Hostname of this target. This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed. An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry).
Name	Optional. Name of this target. Naming the target might be necessary for non digital targets like a human, a vehicle, a building, etc.
Category	Optional. Target category Enumeration available here Table 15
Note	Optional. Free text human-readable additional note for this target.
Service	Optional. Should be High Level Service (messaging, internet, ...), if not defined can be process impacted by the incident.
User	Optional. User ID or login impacted by the incident.
Email	Optional. Email address impacted by the incident. E.g. the value of the "To" header inside a phishing e-mail.
Port	Optional. Target ports involved in the incident. Values in this list MUST be integers and MUST be in the range 1-65535.
Polygon	Optional. Pollygonal incident affected area. Whitespace-delimited list of WGS 84 'latitude,longitude' pairs. Minimum 4 pairs, first equals last
Circle	Optional. Circular incident infected area. WGS 84 center point 'latitude,longitude' followed by a space and radius in kilometers
GeoLocation	Optional. GPS coordinates for the target.
UnLocation	Optional. Standard UN/Locode for the target.
Location	Optional. Internal name for the location of the target.
Locator	Optional. Resolvable URL for the entity. May be dynamic or expire; complements Location, which is a persistent path.
Attachment	Optional. Identifiers for attachments related to this target. Each identifier listed here MUST match the "Name" attribute for one of the attachments described using the Attachment class (Section 5.7).

5.7. The Attachment Class

The Attachment class contains additional data which was captured in relation with the event.

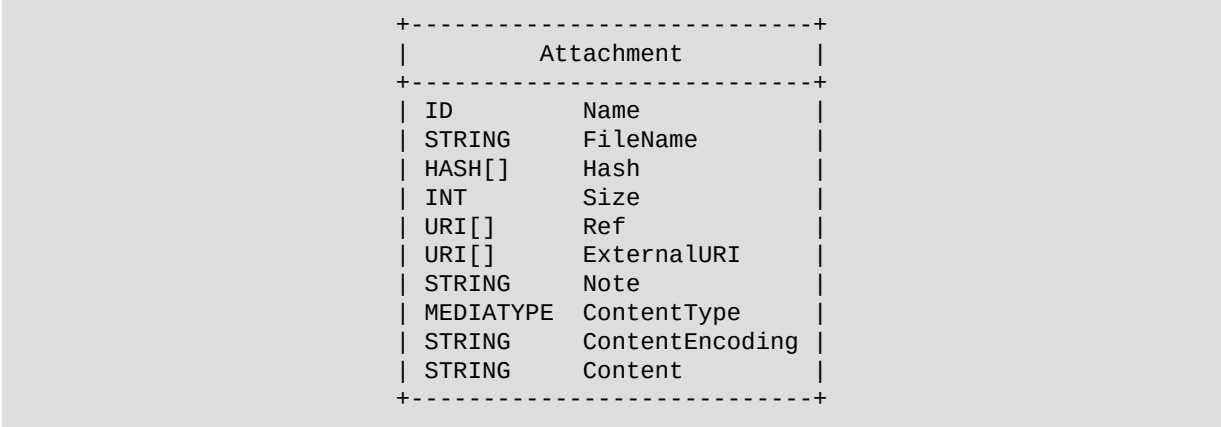


Figure 9: The Attachment class

The Attachment class has the following attributes:

Name	Mandatory. A unique identifier among attachments that can be used to reference this attachment from other classes using the "Attachment" attribute.
FileName	Optional. Attachment filename. This will usually be the original name of the captured file or the name of the file containing the captured content (e.g. a packet capture file).
Hash	Optional. A list of hash results for the attachment's Content. The values in this list are computed by taking the raw value of the attachment's "Content" attribute. The hash result is computed before any other transformation (e.g. Base64 encoding) is applied to the content, so that a receiving IDMEF system may reverse the transformation, apply the same hashing function and obtain the same hash result. See also the definition for the "ContentEncoding" attribute below. It is RECOMMENDED that compatible implementations use one of the hashing functions from the SHA-2 [RFC6234] or SHA-3 [NIST.FIPS.202] families to compute the hash results in this list.
Size	Optional. Length of the content (in bytes). This value MUST be a non-negative integer.
Ref	Optional. References to sources of information related to the incident and/or vulnerability, and specific to this attachment.
ExternalURI	Optional. If the attachment's content is available and/or recognizable from an external resource, this is the URI (usually a URL) to that resource. This MAY also be a URN in a registered or unregistered ad-hoc namespace bearing reasonable information value and uniqueness, such as "urn:mhr:55eaf7effadc07f866d1eaed9c64e7ee49fe081a" or "magnet:?xt=urn:sha1:YNCKHTQCWBTRNJIV4WNAE52SJUQCZO5C".
Note	Optional. Free text human-readable additional note for this attachment.
ContentType	Optional. Internet Media Type of the attachment. For compatibility reasons, implementations SHOULD prefer one of the well-known media types registered in IANA .

ContentEncoding	Optional. Content encoding. The following encodings are defined in this version of the specification: "json": The content refers to a JSON object which has been serialized to a string using the serialization procedure defined in [RFC8259]. "base64": The content has been serialized using the Base64 encoding defined in [RFC4648]. The "base64" encoding SHOULD be used when the content contains binary data. If omitted, the "json" encoding MUST be assumed.
Content	Optional. The attachment's content, in case it is directly embedded inside the message. For large attachments, it is RECOMMENDED that implementations make use of the "ExternalURI" attribute to refererence a copy of the content saved in an external storage mechanism.

5.8. The JavaScript Object Notation Serialization Method

This serialization method aims to convert IDMEFv2 messages to a format that is easy to parse and process, both by software/hardware processors, as well as humans. It relies on the the JavaScript Object Notation (JSON) Data Interchange Format defined in [RFC8259].

Conforming implementations MUST implement all the requirements specified in [RFC8259].

In addition, the following rules MUST be observed when serializing an IDMEFv2 message:

- The top-level Alert class (Section 5.2) is represented as a JSON object ([RFC8259]). This JSON object is returned to the calling process at the end of the serialization process.
- Aggregate classes are represented as JSON objects and stored as members of the top-level JSON object, using the same name as in the IDMEF data model. E.g. the appears under the name "Analyzer" inside the top-level JSON object.
- Attributes are stored as members of the JSON object representing the class they belong to, using the same name as in the IDMEF data model. E.g. the "Version" attribute from the is stored under the name "Version" inside the top-level JSON object.
- Lists from the IDMEF data model are represented as JSON arrays ([RFC8259]). This also applies to aggregate classes where a list is expected. E.g. the "Sensor" member inside the top-level JSON object contains a list of objects, where each object represents an instance of the .
- The various string-based data types listed in Section 3 are represented as JSON strings ([RFC8259]). Please note that the issues outlined in [RFC8259] regarding strings processing also apply here.
- IDMEF attributes with the "NUMBER" data type are represented as JSON numbers ([RFC8259]).

6. Security Considerations

This document describes a data representation for exchanging security-related information between incident detection system implementations. Although there are no security concerns directly applicable to the format of this data, the data itself may contain security-sensitive information whose confidentiality, integrity, and/or availability may need to be protected.

This suggests that the systems used to collect, transmit, process, and store this data should be protected against unauthorized use and that the data itself should be protected against unauthorized access.

The underlying messaging format and protocol used to exchange instances of the IDMEF MUST provide appropriate guarantees of confidentiality, integrity, and authenticity. The use of a standardized security protocol is encouraged.

The draft-lehmann-idmefv2-https-transport-01.txt document defines the transportation of IDMEF over HTTPs that provides such security.

7. IANA Considerations

This document creates 10 identically structured registries to be managed by IANA:

- Name of the registry group: "Incident Detection Message Exchange Format v2 (IDMEF)"
- URL of the registry: http://www.iana.org/assignments/idmefv2
- Namespace format: A registry entry consists of:
  - Rank. A uniq integer for this namespace. Range starts at 0 and ends at the length of this list. The maximum length of this list is 255.
  - Keyword. A keyword for a given IDMEF attribute. It MUST conform to the formatting specified by the IDMEF "ENUM" data type (Section 3.3.1).
  - Description. A short description of the enumerated keyword.
  - Reference. An optional list of URIs to further describe the value.
- Allocation policy: Expert Review per [RFC8126]. This reviewer will ensure that the requested registry entry conforms to the prescribed formatting. The reviewer will also ensure that the entry is an appropriate value for the attribute per the information model (Section 5).

The registries to be created are named in the "Registry Name" column of Table 5. Each registry is initially populated with ranks, keywords and descriptions that come from an attribute specified in the IDMEF model (Section 5). The initial Ranks, Keywords and Description fields of a given registry are listed in "Initial Values". The "Initial Values" column points to a table in this document that lists and describes each enumerated keyword. Each enumerated keyword in the table gets a corresponding entry in a given registry. The initial value of the Reference field of every registry entry described below should be this document.

Registry Name	**Initial Values**
Alert-Type	Table 1 (Alert class (Section 5.2))
Alert-Category	(Alert class (Section 5.2))
Alert-Cause	Table 2 (Alert class (Section 5.2))
Alert-Priority	Table 4 (Alert class (Section 5.2))
Alert-Status	Table 3 (Alert class (Section 5.2))
Analyzer-Category	(Alert class (Section 5.2))
Analyzer-Data	(Analyzer class (Section 5.3))
Analyzer-Method	(Analyzer class (Section 5.3))

Table 5: IANA Enumerated Value Registries

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- The Digital Europe Safe4Soc project (Standard Alert Exchange Format for SOC) who activaly contribute to draft evolution since V04.
- All other european research project having deployed and tested IDMEFv2 on their pilots (PRECINCT, CyberNEMO, CyberSEAS, ATLANTIS, ENDURANCE, KINAITICS, TESTUDO)
- The CESNET team for their work on the [IDEA0] format (based on IDMEFv1) which inspired multiples concepts to IDMEFv2.

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Appendix A. Using IDMEFv2 in Dynamic Environments

1.0

Proposal

October 2026

A.1. The Dynamic Asset Problem

Modern container orchestration platforms and cloud runtime environments create and terminate resources dynamically. IP addresses may change or be reassigned, resource names may be reused, and an entity involved in a security incident may no longer exist when an investigation takes place.

Consequently, identifying a resource solely by its IP address or hostname is insufficient for reliable incident correlation and post-incident investigation.

IDMEFv2 provides complementary attributes that can describe dynamic entities in the , , , and classes. The attribute identifies an entity in the IDMEFv2 data model, while , , and provide additional information about its network presence, operational hostname, and infrastructure context.

These attributes serve different purposes and SHOULD be used together when the relevant information is available. They do not, by themselves, guarantee the long-term availability of historical records or the continued existence of the resource.

A.2. Complementary Identification Attributes

The following attributes have complementary roles when describing dynamic resources.

A.2.1. ID

The attribute identifies an entity in the IDMEFv2 data model. It MUST be populated in accordance with the applicable IDMEFv2 specification.

An IDMEFv2 identifier and a platform-native identifier are different concepts. For example, a Kubernetes pod UID identifies a Kubernetes object instance; it is not a substitute for the IDMEFv2 attribute. Implementations that need to correlate multiple alerts concerning the same resource SHOULD define a consistent mapping between the resource identity and the IDMEFv2 entity identifier. They SHOULD also distinguish successive resource instances when a platform reuses an operational name.

The identifier MUST NOT be assumed to provide access to a resource or to its historical metadata after termination.

A.2.2. IP

The attribute records the network address associated with the entity in the context of the event.

It supports correlation with network telemetry, firewall logs, flow records, and other security monitoring data. However, an IP address is a time-dependent network attribute and MUST NOT be treated as a persistent identifier of an dynamic resource.

When available, the event timestamp and relevant network context SHOULD be retained to support the interpretation of the address at the time of the incident.

Implementations SHOULD distinguish addresses belonging to different infrastructure layers, such as a pod, container, node, NAT gateway, or proxy, when this distinction is relevant to the investigation.

A.2.3. Hostname

The attribute records the hostname observed or configured for the entity. It SHOULD contain an actual hostname, not an unrelated platform identifier.

For a Kubernetes pod, the hostname can be configured through the pod's hostname settings or obtained from the runtime environment. A pod name and the hostname used inside the pod may happen to be the same, but they are not interchangeable by definition. Implementations SHOULD record the actual hostname observed or configured for the pod.

Hostnames can facilitate operational identification and correlation during a resource's lifetime. However, hostnames may be reused, and their uniqueness may be limited to a particular namespace, cluster, runtime, or time interval. Consequently, SHOULD NOT be considered a globally unique or permanently stable identifier without additional context.

A.2.4. Locator

The attribute can carry a platform-specific resource reference that provides additional context for identifying an entity within its infrastructure.

Depending on the environment, this reference MAY contain a cloud resource identifier, an orchestration resource path, a container identifier qualified by its runtime, or a combination of identifiers that uniquely identifies the relevant resource within its scope.

Implementations SHOULD prefer references that identify the individual resource involved in the event rather than only its parent cluster, service, or workload definition.

Where appropriate, a reference MAY include additional context, such as a cluster identifier, namespace, node identifier, container identifier, or resource UID.

Implementations using platform-specific references SHOULD document their syntax, scope, uniqueness properties, and interpretation.

A value does not, by itself, guarantee that a terminated resource can be retrieved or that its historical metadata remains available. Systems requiring post-incident traceability MUST retain the relevant historical metadata and security telemetry for the required investigation period.

A.3. Platform-Specific Resource References

The examples below illustrate how can be populated in different environments. The URI schemes used in these examples, including , , , , and , are illustrative implementation conventions. They are not defined or standardized by IDMEFv2.

The values are illustrative and MUST be replaced with identifiers and hostnames obtained from the relevant platform in an actual deployment. UUID-shaped values shown in the examples are illustrative IDMEFv2 entity identifiers; they are not Kubernetes UIDs unless explicitly placed in a platform reference.

A.3.1. Kubernetes

In Kubernetes, a pod name alone may not be sufficient to distinguish successive instances that share the same name. The pod UID SHOULD be retained when available.

A reference SHOULD include the cluster context, namespace, pod name, and pod UID. Container-specific information MAY be added when the investigation concerns an individual container.

The values below identify IDMEFv2 entities. The values are illustrative actual hostnames configured or observed for the respective pods; they are not merely Kubernetes pod names. The pod names and UIDs are retained in .

Example:

```
{
  "CreateTime": "2026-10-19T10:15:30Z",
  "Analyzer": [
    {
      "ID": "7d6a3c21-1b4e-4f80-9a12-6e5b2c8d901f",
      "IP": "10.10.0.11",
      "Hostname": "secops-collector-01.example.net",
      "Locator": "k8s://cluster-secops/namespace-monitoring/pod/idmefv2-
analyzer?uid=11111111-1111-4111-8111-111111111111"
    }
  ],
  "Sensor": [
    {
      "ID": "8e7b4d32-2c5f-4a91-8b23-7f6c3d9e012a",
      "IP": "10.10.0.12",
      "Hostname": "network-sensor-01.example.net",
      "Locator": "k8s://cluster-prod-eu/namespace-monitoring/pod/network-
sensor?uid=22222222-2222-4222-8222-222222222222"
    }
  ],
  "Source": [
    {
      "ID": "9f8c5e43-3d60-4ba2-9c34-8a7d4e0f123b",
      "IP": "10.244.2.15",
      "Hostname": "payment-worker-01.namespace-paiement.example.net",
      "Locator": "k8s://cluster-prod-eu/namespace-paiement/
pod/payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-
a521-8b7d9c2e104f&container=worker&node=node-worker-03"
    }
  ],
  "Target": [
    {
      "ID": "a09d6f54-4e71-4cb3-8d45-9b8e5f10234c",
      "IP": "10.244.3.20",
      "Hostname": "payment-api-01.namespace-paiement.example.net",
      "Locator": "k8s://cluster-prod-eu/namespace-paiement/pod/payment-
api-7b8c9d-xyz789?
uid=33333333-3333-4333-8333-333333333333&container=api&node=node-
worker-04",
      "Service": "finance"
    }
  ]
}
```

The reference is an illustrative convention. The pod UID identifies the Kubernetes object instance, while the cluster and namespace provide the context required to interpret the reference.

Node-level information MAY be included when it helps correlate the event with network sensors or host-level telemetry.

A.3.2. Docker and Podman

For standalone container runtimes such as Docker and Podman, a container name alone may be insufficient because names can be reused after a container is removed.

SHOULD identify the container instance and, where necessary, the runtime host or engine. The full container identifier SHOULD be preferred over the human-readable container name.

Example:

```
{
  "Target": [
    {
      "ID": "b10e7a65-5f82-4dc4-9e56-0c9f6a21345d",
      "IP": "172.17.0.3",
      "Hostname": "web-app-01.example.net",
      "Locator": "docker://server-web-01/containers/
a1b2c3d4e5f67890123456789012345678901234567890123456789012345678",
      "Service": "e-commerce"
    }
  ]
}
```

The scheme is an illustrative convention, not an official Docker URI scheme. A corresponding convention MAY be defined for Podman, for example using a scheme. Implementations SHOULD document whether the reference identifies the container, the runtime instance, the host, or a combination of these entities.

Where multiple runtime hosts or engines are involved, the reference SHOULD include enough context to avoid ambiguity.

A.3.3. Google Cloud GKE

For Google Kubernetes Engine (GKE), a reference can combine the Google Cloud project and cluster context with the Kubernetes namespace, pod name, and pod UID.

The cluster resource identifier alone identifies the cluster, not the individual pod involved in the incident.

Example:

```
{
  "Target": [
    {
      "ID": "c21f8b76-6083-4ed5-af67-1d0a7b32456e",
      "IP": "10.244.2.15",
      "Hostname": "payment-worker-01.namespace-paiement.example.net",
      "Locator": "gke://projects/my-project-123/locations/us-
west1/clusters/cluster-prod-eu/namespaces/namespace-paiement/
pods/payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-
a521-8b7d9c2e104f&container=worker",
      "Service": "finance"
    }
  ]
}
```

The scheme is an illustrative convention rather than an official Google Cloud resource URI format. Implementations MAY instead use a documented Google Cloud resource identifier together with the Kubernetes identifiers required to identify the individual pod.

A.3.4. AWS ECS

In Amazon Elastic Container Service (ECS), the task ARN identifies a task instance, whereas the task definition ARN identifies a versioned task definition.

For incident investigation involving a particular running or terminated task, SHOULD identify the task instance rather than only its task definition. The task definition ARN MAY be retained separately as supplementary configuration context.

Example:

```
{
  "Target": [
    {
      "ID": "d32f8a76-4b19-4c52-9d63-0a8e7f6b1234",
      "IP": "10.0.1.42",
      "Hostname": "payment-worker-01.example.net",
      "Locator": "arn:aws:ecs:us-west-1:123456789012:task/cluster-prod-
eu/0123456789abcdef0123456789abcdef",
      "Service": "finance"
    }
  ]
}
```

The ARN is illustrative. Implementations MUST use the actual task ARN returned by ECS and SHOULD retain the cluster context when needed to interpret the reference.

When the incident concerns an individual container within a task, the relevant container identifier SHOULD also be retained when available.

The task definition ARN MAY be recorded as additional information to identify the configuration used by the task. It MUST NOT be substituted for the task instance identifier when the objective is to distinguish individual task executions.

A.3.5. Microsoft Azure AKS

In Azure Kubernetes Service (AKS), the Azure Resource Manager (ARM) resource ID of a managed cluster identifies the cluster, not an individual pod.

A pod-level reference SHOULD therefore combine the Azure subscription, resource group, and cluster context with the Kubernetes namespace, pod name, and pod UID.

Example:

```
{
  "Target": [
    {
      "ID": "e43f9b87-5c20-4d63-8e74-1b9f6a20345d",
      "IP": "10.240.0.15",
      "Hostname": "payment-worker-01.namespace-paiement.example.net",
      "Locator": "aks://
subscriptions/12345678-1234-1234-1234-123456789012/resourceGroups/my-
rg/managedClusters/cluster-prod-eu/namespaces/namespace-paiement/pods/
payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-a521-8b7d9c2e104f",
      "Service": "finance"
    }
  ]
}
```

The scheme is an illustrative convention, not an official ARM resource identifier for a pod.

Implementations MAY retain the cluster ARM resource ID as a separate reference or incorporate it into a documented platform-specific representation. The reference SHOULD distinguish the individual Kubernetes resource from its parent cluster.

A.3.6. HashiCorp Nomad

In HashiCorp Nomad, an allocation identifier can provide a reference to a particular workload allocation. The reference SHOULD include sufficient context to distinguish allocations across clusters, regions, namespaces, or jobs, as applicable.

Example:

```
{
  "Target": [
    {
      "ID": "f54a0c98-6d31-4e75-9a82-2c7f1b60345e",
      "IP": "172.17.0.5",
      "Hostname": "payment-worker-01.example.net",
      "Locator": "nomad://region-eu/cluster-prod/job-payment/
alloc-01234567-89ab-cdef-0123-456789abcdef?task=worker",
      "Service": "finance"
    }
  ]
}
```

The scheme is an illustrative convention. Implementations SHOULD use the actual allocation identifier provided by Nomad and document the context required to interpret it.

Task-level or container-level information MAY be included when necessary to distinguish the specific runtime entity involved in the event.

A.4. Scope of the Service Attribute

The attribute in the class SHOULD describe the high-level service or process affected, consistently with the IDMEFv2 specification.

For example, values such as , , and can identify business-level services associated with the target.

Implementations SHOULD avoid using as an undocumented substitute for platform-specific resource identifiers, such as pod names, container IDs, allocation IDs, task ARNs, or cluster resource IDs.

Where business services and technical runtime services must be distinguished, implementations SHOULD document the mapping between these concepts and the relevant IDMEFv2 attributes.

The use of for a particular purpose MUST remain consistent with the attribute definition in the applicable version of the IDMEFv2 specification.

A.5. Temporal Context and Historical Retention

Reliable post-incident investigation requires more than a stable resource reference.

Implementations SHOULD preserve the event timestamp and sufficient contextual information to determine which resource instance, network address, and runtime environment were involved at the time of the event.

The IDMEFv2 time attributes SHOULD be used according to their defined semantics. In particular, indicates when the alert was created, whereas describes the start of the activity or situation being reported when that information is available. They are not interchangeable, and neither should be assumed to represent the exact measurement time of every individual attribute. Implementations SHOULD retain any additional timestamps required to interpret time-sensitive measurements accurately.

Systems requiring long-term traceability MUST retain the historical metadata and security telemetry needed for the intended investigation period. Such information can include orchestration events, container lifecycle records, cloud audit logs, network flow records, and mappings between resource identifiers and network addresses.

A value MAY be used as a lookup key into retained historical records. However, the availability and retention of those records are properties of the surrounding monitoring and logging infrastructure, not guarantees provided by the attribute itself.

A.6. Summary

Ephemeral environments require a distinction between entity identification, network correlation, and historical resource context.

The attribute identifies the entity in the IDMEFv2 data model. records the network address observed in the context of the event, provides the actual configured or observed hostname, and can carry a platform-specific reference to the relevant resource.

Implementations SHOULD use resource references that identify individual instances rather than only their parent clusters or configuration definitions. They SHOULD preserve sufficient temporal and infrastructure context to distinguish successive instances and correlate events across security telemetry sources.

Long-term traceability additionally depends on the retention and availability of historical metadata and security records.

This approach enables IDMEFv2 to represent dynamic resources across different orchestration platforms and container runtimes without requiring a separate IDMEFv2 extension for each environment. Platform-specific reference formats remain implementation conventions and SHOULD be documented to support interoperability.

The JSON fragments are illustrative. Their exact structure MUST conform to the applicable IDMEFv2 serialization and schema.

Appendix B. IDMEFv2 Alert Signature Specification

1.0

Proposal

October 2026

This appendix defines a standardized mechanism for signing IDMEFv2 alerts. The signature ensures the integrity and authenticity of alert data while allowing certain attributes to remain modifiable without invalidating the signature (partial signing).

The mechanism builds upon the following standards:

JWS (RFC 7515) — signature format;

JCS (RFC 8785) — JSON canonicalization;

JWA (RFC 7518) — cryptographic algorithms;

JWK (RFC 7517) — key representation;

IDMEFv2 Paths (Section 3.3.12 of the draft) — attribute selection.

The key words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, **MAY** and **RECOMMENDED** are to be interpreted as described in RFC 2119.

B.1. Overview

The mechanism adds two optional attributes to the class:

1. ``**SignedAttributes**``: a list of IDMEFv2 paths identifying the attributes to be signed;
2. ``**Signature**``: a string containing the JWS Compact Serialization of the signature.

The signing process follows these steps:

1. populate with the list of IDMEFv2 paths;
2. build the signed payload object as defined in Signed Attributes and Payload Construction (Appendix B.3);
3. canonicalize this object using JCS (RFC 8785);
4. build the protected JWS header (containing , and optionally);
5. produce the JWS Compact Serialization as defined in Signature Format: JWS Compact Serialization;
6. store the result in the attribute.

B.2. Signature Format: JWS Compact Serialization

The attribute **MUST** use the JWS Compact Serialization format defined in RFC 7515.

B.2.1. Structure

```
BASE64URL(JWS Protected Header) "." BASE64URL(JWS Payload) "."
 BASE64URL(JWS Signature)
```

The JWS payload **MUST** be **embedded** (not detached) in the Compact serialization. It corresponds to the canonicalized representation of the payload object defined in Signed Attributes and Payload Construction (Appendix B.3), encoded in base64url as specified by RFC 7515.

> *Note: The payload is derived from the alert itself; embedding it guarantees that the verifier has access to the exact representation that was signed, independently of any re-serialization of the alert. The unencoded payload option of RFC 7797 is **not** used.*

B.2.2. Protected Header

The protected JWS header **MUST** contain the parameter. It **SHOULD** contain the parameter. It **MUST NOT** contain any parameter not specified by this annex.

B.2.3. Example

```
"Signature": "<valid JWS Compact string, to be provided in test vectors>"
```

B.3. Signed Attributes and Payload Construction

B.3.1. Path Syntax

Attributes to be signed are identified using the IDMEFv2 path syntax defined in Section 3.3.12 of the draft.

Attribute type	Example
Top-level attribute	
Nested class attribute	
List attribute (first element)	(index 0 implied)
List attribute (specific index)	

Table 6

B.3.2. Payload Structure (Normative)

The signed payload **MUST** be a **JSON object** whose structure mirrors the nesting of the paths listed in . Specifically:

- a top-level path produces a key at the root;
- a nested path produces an object containing a key ;
- a list path (or with an implied index 0) produces an **array** under the key , whose element at index 0 is an object containing the key ;
- list indices **MUST** be contiguous starting from 0 in the payload; unreferenced positions **MUST NOT** be included;
- **MUST** be included in the payload under the key , with the list of paths as provided (see Constraints on SignedAttributes (Appendix B.3.4));
- **MUST NOT** be included in the payload.

Example. For , the payload is:

```
{
  "Version": "2.D.V08",
  "ID": "123e4567-e89b-12d3-a456-426614174000",
  "Analyzer": { "Name": "Corporate EDR" },
  "Source": [ { "IP": "198.51.100.20" } ],
  "SignedAttributes": ["Version", "ID", "Analyzer.Name", "Source(0).IP"]
}
```

B.3.3. Absent or Invalid Attributes

If a path listed in does not correspond to any attribute present in the alert, the producer **MUST** reject the signing operation (validation error). A verifier that observes the absence of a referenced attribute **MUST** reject the signature.

B.3.4. Constraints on SignedAttributes

- **MUST NOT** contain or .

- The order of paths in the list is **not significant** (see Attribute Order: Not Significant (Appendix B.5)).
- The list **MUST** be included in the payload to bind the signature to the covered set of paths.

B.4. Attributes Added to the Alert Class

Attribute	Type	Description
		List of IDMEFv2 paths of the attributes included in the signature.
		JWS Compact Serialization string containing the signature.

Table 7

B.4.1. SignedAttributes

Optional. **MUST** be present when is present. **MUST** be included in the JWS payload (see Payload Structure (Normative) (Appendix B.3.2)).

B.4.2. Signature

Optional. Present only when the alert is signed. Contains the complete JWS Compact Serialization.

B.4.3. Consistency

An alert that contains without , or vice versa, **MUST** be considered invalid by a verifier.

B.5. Attribute Order: Not Significant

The order of attributes in does not affect the signature because:

1. the payload is canonicalized using JCS (RFC 8785);
2. JCS deterministically sorts all keys in lexicographic order;
3. the same logical object therefore always produces the same canonical representation.

Note: The array in the payload preserves the order of the specified indices; JCS does not reorder array elements.

B.6. Normalization: JSON Canonicalization Scheme (JCS)

All JSON objects used as the JWS payload **MUST** be canonicalized using JCS (RFC 8785) before signing and before verification.

B.6.1. JCS Rules

- **o**bject keys are sorted lexicographically;
- **w**hitespace outside string literals is removed;
- **n**umbers are formatted according to I-JSON rules (RFC 7493);
- **;** and are forbidden.

B.6.2. Rationale

Without canonicalization, the same logical JSON object could be represented differently (key ordering, whitespace), leading to signature verification failures. JCS ensures deterministic serialization.

B.7. Signature Algorithms

B.7.1. Supported Algorithms

The following algorithms **MUST** be supported:

Algorithm	Type	Usage
	ECDSA P-256 / SHA-256	RECOMMENDED
	RSASSA-PSS / SHA-256	RECOMMENDED
(Ed25519)	EdDSA curve 25519	RECOMMENDED
	RSA PKCS#1 v1.5 / SHA-256	IF NECESSARY

Table 8

B.7.2. Excluded Algorithms

The following algorithms **MUST NOT** be used:

(unsecured);

; , (HMAC — inappropriate for third-party verification);

any algorithm not listed in JWA (RFC 7518).

B.7.3. Verification Policy

A verifier **MUST** check that the value of the protected header corresponds to an algorithm allowed by its local policy, and **MUST** reject the signature otherwise (protection against algorithm confusion).

B.8. Key Management

B.8.1. Key Identification

``alg`` parameter. MUST be included in the protected JWS header. It identifies the cryptographic algorithm used to secure the JWS. The signature is not valid if does not represent a supported algorithm, or if no key associated with the signing party is available for that algorithm. The value **MUST** be a case-sensitive string containing a registered algorithm name (from the IANA "JSON Web Signature and Encryption Algorithms" registry as defined in JWA) or a value that contains a Collision-Resistant Name.

``kid`` parameter. SHOULD be included in the protected JWS header. It provides a hint indicating which key was used to secure the JWS, allowing the recipient to locate the verification key and facilitating key rotation. Other identification mechanisms (out-of-band agreement) **MAY** be used, which is why this parameter remains optional.

Example header:

```
{ "alg": "ES256", "kid": "key-2025-01" }
```

B.8.2. Key Representation

Keys **SHOULD** be represented in JWK format (RFC 7517) when exchanged as part of the payload. Other mechanisms, such as referencing a key via a understood within a pre-existing PKI, **MAY** be used to maintain compatibility with existing infrastructures.

B.8.3. Key Distribution

The key distribution mechanism is outside the scope of this annex. The following methods **MAY** be used: pre-shared public keys, PKI / X.509 certificates, OAuth 2.0 or OpenID Connect metadata endpoints, manual exchange.

B.9. Verification Procedure

A verifier **MUST** apply the following steps:

1. check for the joint presence of and (Consistency (Appendix B.4.3)); reject otherwise;
2. decode the JWS Compact serialization and check that it has exactly three segments;
3. check that the protected header contains and that its value is allowed by the local policy (Verification Policy (Appendix B.7.3)); reject otherwise;
4. locate the verification key using or another mechanism (Key Management (Appendix B.8));
5. rebuild the payload object from the alert and as defined in Payload Structure (Normative) (Appendix B.3.2); if a referenced attribute is absent, reject (Absent or Invalid Attributes (Appendix B.3.3));
6. canonicalize this object with JCS and compare the result with the decoded payload; reject on any difference;
7. verify the cryptographic signature using the identified key and algorithm; reject on failure;
8. on success, consider as intact and authentic only the attributes listed in .

   Note: An alert whose and have been removed becomes an unsigned alert. Recipients **SHOULD** apply an explicit policy (rejection or marking) to unsigned alerts when a signature is expected.

B.10. Security Considerations

B.10.1. Key Protection

Private keys **MUST** be protected against unauthorized access. Implementations **SHOULD** use hardware security modules (HSM) or secure enclaves for key storage.

B.10.2. Algorithm Agility

To ensure the long-term security of IDMEFv2 alerts, implementations **SHOULD** support algorithm agility, allowing migration to newer algorithms as cryptographic best practices evolve. As post-quantum signature algorithms (for example ML-DSA, SLH-DSA) are standardized and become widely available, they **MAY** be used in place of, or in combination with, classical algorithms (for example via hybrid signatures). The parameter facilitates this transition by enabling different key types to be identified without changing the IDMEFv2 data model.

B.10.3. Replay Attacks

JWS does not inherently prevent replay attacks. Implementations **SHOULD** include time-based attributes among the signed attributes (for example) and/or rely on transport-layer security. The use of JWT-like / claims is **outside the scope** of the IDMEFv2 model and **MUST NOT** be assumed by a generic verifier.

B.10.4. Key Rotation

Regular key rotation **SHOULD** be implemented. The parameter facilitates it by allowing the recipient to identify the key used.

B.11. Complete Example

```
{
  "Version": "2.D.V08",
  "ID": "123e4567-e89b-12d3-a456-426614174000",
  "CreateTime": "2024-05-21T14:30:00Z",
  "Type": "Cyber",
  "Category": "Access.Compromise",
  "Confidence": 0.95,
  "Analyzer": {
    "ID": "analyzer01",
    "Name": "Corporate EDR",
    "IP": "192.0.2.10"
  },
  "Source": [ { "IP": "198.51.100.20" } ],
  "SignedAttributes": [
    "Version", "ID", "CreateTime", "Type", "Category",
    "Analyzer.ID", "Analyzer.Name", "Source(0).IP"
  ],
  "Signature": "<valid JWS Compact string>"
}
```

The corresponding canonicalized payload (before base64url encoding) is:

```
{"Analyzer":{"ID":"analyzer01","Name":"Corporate
 EDR"},"Category":"Access.Compromise","CreateTime":"2024-05-21T14:30:00Z","ID":"123e456
e89b-12d3-a456-426614174000","SignedAttributes":
["Version","ID","CreateTime","Type","Category","Analyzer.ID","Analyzer.Name","Source(0)
[{"IP":"198.51.100.20"}],"Type":"Cyber","Version":"2.D.V08"}


Note: ``Confidence`` and ``Analyzer.IP`` do not appear in the payload
 because they
are not listed in ``SignedAttributes``. Signed and verifiable test vectors
**MUST** accompany the final version.
```

B.12. Conformance and Test Vectors

An implementation is conformant with this annex if it satisfies all the **MUST** and **MUST NOT** requirements above. The final version **MUST** provide:

- at least one test vector per supported algorithm (Supported Algorithms (Appendix B.7.1));
- one negative test vector per rejection case (Absent or Invalid Attributes (Appendix B.3.3), Consistency (Appendix B.4.3), Verification Policy (Appendix B.7.3), Verification Procedure (Appendix B.9));
- the expected canonicalized payload and signature values.

B.13. References

B.13.1. Normative References

- RFC 2119 — Key words for use in RFCs to Indicate Requirement Levels
- RFC 7515 — JSON Web Signature (JWS)
- RFC 7517 — JSON Web Key (JWK)
- RFC 7518 — JSON Web Algorithms (JWA)

- RFC 7493 — The I-JSON Message Format
- RFC 8259 — The JavaScript Object Notation (JSON) Data Interchange Format
- RFC 8785 — JSON Canonicalization Scheme (JCS)
- IDMEFv2 Draft — draft-lehmann-idmefv2-08 (Section 3.3.12)

B.13.2. Informative References

- RFC 7519 — JSON Web Token (JWT)
- RFC 7797 — JSON Web Signature (JWS) Unencoded Payload Option

Appendix C. Enumerations

C.1. Entity Sector

Rank	Keyword	Definition
0	Undefined	Sector undefined
1	Administration.CentralGovernment	Central government ministries and agencies
2	Administration.Diplomatic	Embassies, consulates, and diplomatic missions
3	Administration.EmergencyServices	Police, fire, rescue, and emergency medical services
4	Administration.Judiciary	Courts, judicial systems, and correctional facilities
5	Administration.Local	Regional and municipal public services
6	Administration.Other	Other public administration entities critical for the continuity of public services
7	Banking.Banking	Banking institutions and credit activities
8	Banking.Other	Other banking and financial entities critical for the stability of the financial system
9	Cemeteries.Crematoria	Crematoria operations
10	Cemeteries.Public	Public cemetery and crematorium facilities
11	Cemeteries.Other	Other cemetery and funeral service entities critical for public health and social continuity
12	Chemical.Production	Production of hazardous and high-consequence chemicals
13	Chemical.Storage	Chemical storage and distribution facilities
14	Chemical.Other	Other chemical sector entities critical for safety and security
15	Civil.ElectionMonitoring	Election observation and monitoring entities
16	Civil.HumanitarianAid	Humanitarian aid organizations
17	Civil.NGOs	Non-governmental organizations with critical functions
18	Civil.Other	Other civil society entities critical for democratic processes and social stability
19	Cultural.Archives	National and regional archives
20	Cultural.HistoricalSites	Historical monuments and archaeological sites
21	Cultural.Libraries	National and public libraries
22	Cultural.Museums	Museums and exhibition spaces

Rank	Keyword	Definition
23	Cultural.Other	Other cultural heritage entities critical for national identity and continuity
24	Defense.CommandControl	Military C4ISR systems (command, control, communications, computers, intelligence, surveillance, reconnaissance)
25	Defense.Cyber	Military cyber defense and security operations centers
26	Defense.IndustrialBase	Defense manufacturing, weapons systems, munitions production
27	Defense.Logistics	Military supply chains, fuel depots, ammunition storage
28	Defense.Installations	Military bases, headquarters, and operational facilities
29	Defense.Research	Defense laboratories and R&D facilities
30	Defense.Other	Other defense and military entities critical for national security
31	Digital.CloudServices	Cloud computing and critical digital services
32	Digital.DataCenters	Data center hosting and infrastructure services
33	Digital.DigitalProviders	Online marketplaces, search engines and social media platforms
34	Digital.DomainNameSystems	DNS service providers and TLD registries
35	Digital.ICTServiceManagement	ICT service management B2B services
36	Digital.SatelliteCommunications	Satellite communication networks and ground stations
37	Digital.TelecomNetworks	Fixed and mobile telecommunications networks
38	Digital.UnderseaCables	Submarine communications cable infrastructure
39	Digital.Other	Other digital and telecommunications entities critical for the continuity of digital services
40	Education.PrimarySecondary	Primary and secondary schools
41	Education.ResearchSchools	Research-focused educational institutions
42	Education.Universities	Universities and higher education institutions
43	Education.Other	Other educational entities critical for societal continuity
44	Emergency.CivilProtection	Civil protection and disaster management agencies

Rank	Keyword	Definition
45	Emergency.Medical	Ambulance services and emergency medical response
46	Emergency.FireAndRescue	Firefighting and rescue services
47	Emergency.Police	Law enforcement agencies and operations
48	Emergency.Other	Other emergency response entities critical for public safety
49	Energy.DistrictHeating	Operation of district heating networks
50	Energy.Electricity	Generation, transmission and distribution of electrical power
51	Energy.Gas	Transport, storage and distribution of natural gas
52	Energy.Hydrogen	Production, transport and storage of hydrogen
53	Energy.Nuclear	Nuclear power generation and fuel cycle facilities
54	Energy.Oil	Refining, transport and storage of petroleum products
55	Energy.Other	Other energy entities critical for the continuity of energy supply
56	Finance.MarketInfrastructures	Stock exchanges and clearing houses
57	Finance.Insurance	Systemically important insurance entities
58	Finance.PaymentSystems	Card schemes, transfers and instant payment systems
59	Finance.Other	Other financial entities critical for the stability of the financial system
60	Food.AgriculturalProduction	Strategic agricultural and livestock production
61	Food.FoodDistribution	Retail and logistics for food supply chains
62	Food.FoodProcessing	Industrial food processing and manufacturing
63	Food.FoodSafety	Food safety inspection and control authorities
64	Food.Other	Other food sector entities critical for the continuity of the food supply chain
65	Health.BloodAndTissue	Blood banks, tissue banks, and transplantation services
66	Health.HealthcareProviders	Hospitals and clinics providing essential care
67	Health.Laboratories	Medical analysis and diagnostic laboratories
68	Health.MedicalDevices	Manufacture and maintenance of critical medical devices
69	Health.PharmaceuticalSupplyChain	Manufacturing, wholesale and distribution of pharmaceuticals
70	Health.PublicHealth	Public health agencies and epidemiological surveillance

Rank	Keyword	Definition
71	Health.Other	Other health entities critical for the continuity of healthcare services
72	Logistics.FreightForwarding	Freight forwarding and cargo management
73	Logistics.LastMileDelivery	Last-mile delivery services
74	Logistics.ThirdPartyLogistics	Third-party logistics providers
75	Logistics.Warehousing	Strategic warehousing and storage facilities
76	Logistics.Other	Other logistics entities critical for supply chain continuity
77	Manufacturing.Aerospace	Aerospace and aviation manufacturing
78	Manufacturing.Batteries	Battery manufacturing and energy storage production
79	Manufacturing.Chemical	Chemical manufacturing and industrial processes
80	Manufacturing.Electronic	Manufacture of computer, electronic and optical products
81	Manufacturing.Defense	Defense manufacturing and strategic military supply chains
82	Manufacturing.MedicalDevices	Manufacture of medical devices and equipment
83	Manufacturing.MotorVehicles	Manufacture of motor vehicles and transport equipment
84	Manufacturing.Pharmaceutical	Pharmaceutical manufacturing and active pharmaceutical ingredients
85	Manufacturing.Semiconductors	Semiconductor fabrication and microelectronics manufacturing
86	Manufacturing.Strategic	Critical manufacturing such as metallurgy and components
87	Manufacturing.Other	Other manufacturing entities critical for industrial resilience and strategic supply chains
88	Media.Infrastructure	Transmitters, towers, and broadcast distribution infrastructure
89	Media.OnlineMedia	Digital news platforms and content providers
90	Media.Press	Newspapers, publishing houses, and press agencies
91	Media.Radio	Radio broadcasting networks and studios
92	Media.Television	Television broadcasting networks and studios
93	Media.Other	Other media entities critical for information integrity and public communication
94	Mining.CriticalRawMaterials	Extraction and processing of critical raw materials
95	Mining.EnergyMinerals	Coal, uranium, and other energy mineral mining

Rank	Keyword	Definition
96	Mining.Other	Other mining entities critical for resource security
97	Nuclear.FuelCycle	Nuclear fuel production, enrichment, reprocessing, and waste management
98	Nuclear.Medical	Nuclear medical facilities and radioisotope production
99	Nuclear.PowerGeneration	Civil nuclear power plants and associated facilities
100	Nuclear.Research	Nuclear research reactors and laboratories
101	Nuclear.Other	Other nuclear entities critical for safety and security
102	Postal.CourierServices	Postal and courier services
103	Postal.Other	Other postal and courier entities critical for the continuity of mail and parcel services
104	Religious.PilgrimageSites	Major pilgrimage destinations
105	Religious.PlacesOfWorship	Churches, mosques, synagogues, temples, and other religious buildings
106	Religious.Other	Other religious sites critical for community continuity
107	Research.BiologicalSafety	BSL-3 and BSL-4 high-containment laboratories
108	Research.ChemicalSafety	High-containment chemical research facilities
109	Research.Research	Key research laboratories with strategic importance
110	Research.Other	Other research entities critical for strategic research continuity and innovation
111	Space.GroundStations	Satellite ground control and telemetry stations
112	Space.LaunchFacilities	Space launch sites and associated infrastructure
113	Space.SpaceActivities	Space industry and satellite operations
114	Space.Other	Other space entities critical for the continuity of space-based services and infrastructure
115	Transport.Aviation	Airports, air traffic control and airline operations
116	Transport.Maritime	Ports, terminals and maritime traffic management
117	Transport.Pipeline	Oil, gas, and hydrogen pipeline infrastructure
118	Transport.PublicTransport	Urban and regional public transportation systems
119	Transport.Rail	Railway infrastructure and train operations
120	Transport.Road	Traffic management and strategic road logistics

Rank	Keyword	Definition
121	Transport.Other	Other transport entities critical for the continuity of passenger and freight mobility
122	Waste.Hazardous	Collection, treatment and disposal of hazardous waste
123	Waste.NonHazardous	Management of non-hazardous solid waste
124	Waste.Nuclear	Nuclear waste management and storage facilities
125	Waste.Recycling	Waste processing and recycling operations
126	Waste.Other	Other waste management entities critical for the continuity of waste services
127	Water.DamsAndReservoirs	Dam and reservoir infrastructure for water management
128	Water.DrinkingWater	Capture, treatment and distribution of potable water
129	Water.Irrigation	Large-scale agricultural irrigation systems
130	Water.Wastewater	Collection and treatment of sewage and wastewater
131	Water.Other	Other water entities critical for the continuity of water supply and sanitation

Table 9: enum_alert_entitysector

C.2. Alert Category

Rank	Keyword	Description
0	Abuse.Coercion	The practice of compelling an individual to act against their will by using force, threats, intimidation, or extreme dependency, often for personal or financial gain. Examples: an insider forced by a third party to disclose credentials, a person pressured into handing over access badges.
1	Abuse.Extremism	The process of socializing an individual, often through psychological manipulation, to adopt the beliefs and goals of a violent extremist group and become willing to engage in activities that support the group. Examples: online radicalization through extremist forums, recruitment within a closed community.

Rank	Keyword	Description
2	Abuse.Grooming	The deliberate process of building an emotional connection and trust with a person to lower their inhibitions and defenses, in order to psychologically manipulate them for purposes such as sexual abuse, exploitation, trafficking, financial fraud, organizational coercion, or illicit insider activity. Ex: an adult building trust with a minor via a gaming platform, with repeated "friendly" contacts near a school; or a colleague cultivating a relationship of dependency to draw someone into fraud or insider dealing.
3	Abuse.Harassment	A pattern of unwanted, intrusive behaviour (physical, verbal, or online) directed at a specific person that causes fear, distress, or emotional harm, or any such behaviour—including a single severe incident—that creates a hostile or intimidating environment. Ex: sustained threatening messages on social media, repeated following of an employee, or a one-off serious act of intimidation.
4	Abuse.Trafficking	The act of recruiting, transporting, transferring, harbouring, or receiving a person through force, fraud, or coercion for the purpose of exploitation, such as forced labour or sexual servitude. Ex: recruitment via fake job ads for forced online scams, transport of victims across a border.
5	Abuse.Other	Any other incident involving manipulation or coercion of people for harmful purposes that does not fit into the specific abuse subcategories. Examples: psychological domination of a remote worker, confinement of a person.
6	Access.Authorized	An incident involving legitimate, approved physical or digital access to resources that

Rank	Keyword	Description
		may be relevant for auditing, monitoring, or establishing a baseline of normal activity. Examples: an admin login during a maintenance window, an employee entering a server room with a valid badge, a packet accepted through a firewall.
7	Access.Backdoor	The installation or discovery of a concealed method of bypassing normal authentication or encryption in a computer system, product, embedded device or a building. Examples: a hidden SSH key installed on a server, a concealed door in a facility.
8	Access.Cloned	The act of creating an unauthorized copy of a physical or digital security credential, such as an access card, to gain illicit entry to a facility or system. Examples: a duplicated RFID badge, a copied key to a locked cabinet.
9	Access.Compromise	An incident where a legitimate user's credentials (e.g., username/password, session tokens, API keys, SSH keys and biometric spoofing) are stolen or guessed and used by an unauthorized individual to gain access to systems or data. Examples: stolen VPN credentials reused by an attacker, a stolen badge used to enter a building.
10	Access.Escalation	An incident where a user or process gains access rights, permissions, or capabilities that exceed those normally assigned, often to bypass security restrictions. Examples: privilege escalation to domain admin, a visitor reaching a restricted floor.
11	Access.Expired	An event indicating that an access attempt was made using credentials, tokens, or a badge whose validity period has ended. The incident may be benign (an oversight) or

Rank	Keyword	Description
		malicious (deliberate reuse of expired credentials to bypass controls). Examples): a session token past its expiration is accepted by an application due to a validation flaw, granting continued access , an individual enters a protected area using an access badge whose authorization expired after their training lapsed .
12	Access.Forced	An incident involving the use of physical force to breach a barrier (e.g., lock, door, window) or the repeated systematic guessing of passwords to gain unauthorized access (brute-force attack). Examples: a brute-force attack on an SSH service, forced entry through a window.
13	Access.Lost	An incident where physical or digital access credentials (e.g., keys, ID badges, login details) are misplaced, stolen, or used by an unauthorized person. Examples: a lost hardware token, a misplaced key ring.
14	Access.Tailgating	A security breach where an unauthorized person follows an authorized individual through a secured entry point, circumventing access control mechanisms. Examples: following an employee through a badge door, session hijacking after an authenticated user.
15	Access.Unauthorized	Incident involving unauthorized access to a restricted building, room, area, system, network, application, account, or data resource, whether actual or attempted. Examples: a connection to a database without rights, entry into a restricted area, packet denied by a firewall.
16	Access.Other	Any other incident related to physical or digital access that does not fit into the specific access subcategories. Examples: access via an

Rank	Keyword	Description
		undocumented API, entry through an unmonitored gate.
17	Anomaly.Burst	A sudden, abrupt, high-amplitude deviation over a short window, whether or not followed by a return to normal. Examples: a sudden spike in connections or requests, a panic movement in a crowd, a sudden gust or storm, a pollution peak.
18	Anomaly.Chaos	A disordered, unpredictable behavior with high entropy, showing no identifiable pattern or stable structure. Examples: erratic network traffic, chaotic crowd movement, atmospheric turbulence, uncontrolled spread of a wildfire.
19	Anomaly.Context	A behavior that is normal in absolute terms but abnormal in its context (location, role, time, system state, threat level). Examples: an admin logging in from an unusual country, a vehicle on a forbidden lane, ordinary activity during a crisis, a normal crowd in a place where it is not expected.
20	Anomaly.Correlation	An unusual covariation between several entities, signals, or domains that should not evolve together, or the breaking of a normally expected correlation. Examples: correlation between VPN access and physical access, between weather and incidents, between network flows and crowd behavior; or conversely, a sensor losing its expected correlation.
21	Anomaly.Deviation	Any measurable deviation from an established behavior, baseline, or norm, without presupposing its temporal shape or amplitude. This is the generic subcategory, to be used when no other applies. Examples: a user account logging in from an unusual location, a crowd whose density exceeds the normal range, an out-of-season temperature.

Rank	Keyword	Description
22	Anomaly.Drift	A slow, gradual, and almost imperceptible deviation that settles over time and only becomes visible when comparing distant time windows. Examples: slow drift of an account's privileges, gradual erosion of an infrastructure, progressive warming, slow change in crowd or traffic composition.
23	Anomaly.Oscillation	A repeated alternation between two or more states, without convergence or stabilisation, over a given window. Examples: flapping of a service, back-and-forth movement of a crowd, sensor oscillation, alternating weather conditions.
24	Anomaly.Relation	A deviation affecting the set, structure, or topology of relations between an asset and other assets, rather than the asset's own value, timing, or sequence. Examples: a workstation that normally connects to a stable set of servers suddenly connects to a new one; an asset whose movement pattern shifts to an entirely new set of locations.
25	Anomaly.Sequence	An anomaly affecting the order, succession, or cadence of normally expected events, rather than their value. Examples: process steps reversed or skipped, unusual movement order in a crowd, atypical weather sequence (frost then out-of-season heatwave).
26	Anomaly.Threshold	Crossing a predefined threshold by a simple, non-statistical rule, without baseline analysis. Examples: a temperature threshold, a crowd density threshold, a connection attempt threshold, a network throughput threshold.
27	Anomaly.Timing	An anomaly affecting the moment when an event occurs, normal in itself but happening at an unusual time. Examples: a login at an atypical hour, a

Rank	Keyword	Description
		crowd arriving too early or too late, out-of-season rain, activity outside expected hours.
28	Anomaly.Unknown	An anomaly that has been detected but cannot be classified into the subcategories above, or whose nature remains undetermined after analysis. Examples: an unclassified deviation flagged by a detection system, an anomaly whose cause is still under investigation.
29	Anomaly.Other	Any other anomaly that does not fit into the specific subcategories. Examples: a deviation that combines several anomaly types without a dominant one.
30	Availability.DDoS	An incident where multiple compromised systems (botnet), assets or individuals are used to target a single system with a flood of traffic, causing a denial of service. Examples: a botnet flooding a web service, saturation of a phone exchange.
31	Availability.DoS	An incident where a single machine or network attempts to make a system, service, or network resource unavailable by overwhelming it with malicious requests or traffic. Examples: a single-host SYN flood, jamming of a radio channel.
32	Availability.Failure	An incident caused by the unintentional malfunction of hardware or software due to errors, bugs, wear and tear, or other faults, leading to service degradation or unavailability. Examples: a disk failure taking down a database, a pump failure stopping water supply, a car breakdown.
33	Availability.Flapping	A pattern of rapid, repeated transitions between operational states (such as up and down) of a system, service, process, or component within a short time window. Examples: a service

Rank	Keyword	Description
		alternating up and down, an intermittent power relay.
34	Availability.Heartbeat	A periodic signal generated by hardware, software or individual to indicate normal operation, often used for monitoring system health, connectivity, or location tracking. Examples: a keep-alive signal from an agent, a periodic beacon from a patrol device.
35	Availability.Misconfiguration	An incident where incorrect configuration of systems, software, or networks leads to service disruptions, outages, or security vulnerabilities. Examples: a wrong firewall rule blocking traffic, a mis-set HVAC control.
36	Availability.Outage	An incident where essential utilities or services (such as electricity, water, or network connectivity) become unavailable, disrupting normal operations. Examples: a cloud region outage, a power cut in a building, a widespread grid failure after a storm.
37	Availability.Overload	An incident where a system or component is subjected to a load beyond its designed capacity, leading to performance degradation or failure, even if the load is not malicious. Examples: CPU saturation on a server, overcapacity of a transport network.
38	Availability.Recovery	The return of a system, service, process, or component to normal operation following a failure, overload, outage, or other disruptive condition. Examples: a service back online after an incident, power restored after an outage, water supply restored after a flood.
39	Availability.Restart	The planned or unplanned reinitialisation of a system, service, process, or component, involving a stop followed by a start. Examples: a planned

Rank	Keyword	Description
		server reboot, a restart of a generator.
40	Availability.Start	The initialisation or startup of a system, service, process, or component, transitioning from an inactive to an active state. Examples: a service startup after deployment, the start of a pump.
41	Availability.Status	A point-in-time report describing the current state, health, or operational metrics of a system, service, or component, typically emitted on demand or at significant transitions. Examples: a health-check report from a node, a status board of a facility, the polling of a normal disk space.
42	Availability.Stop	The intentional or unintentional termination of a system, service, process, or component, transitioning from an active to an inactive state. Examples: a service shutdown, the stop of a production line.
43	Availability.Other	Any other incident that impacts the availability of resources or services, not covered by the specific subcategories. Examples: a DNS resolution failure, a lift out of service.
44	Biological.Animal	An incident where animals pose a direct threat to human safety, public health (e.g., zoonotic diseases), or economic stability (e.g., livestock diseases). Examples: an aggressive dog loose on a site, a zoonotic outbreak from livestock.
45	Biological.Epidemic	The rapid and widespread occurrence of an infectious disease in a specific population or region, exceeding what is normally expected. Examples: a regional flu epidemic, a measles outbreak in a community.
46	Biological.Insect	An incident involving a harmful outbreak or infestation of insects that threatens public health, agriculture, livestock, or property. Examples: a locust

Rank	Keyword	Description
		swarm threatening crops, an invasive mosquito infestation.
47	Biological.Zombies	A fictional or hypothetical scenario involving a pathogen that causes a pandemic of aggressive, infectious behavior, often used as a metaphor for worst-case outbreak scenarios in planning. Examples: a tabletop exercise simulating a zombie pandemic, a worst-case planning scenario for an uncontrolled outbreak.
48	Biological.Other	Any other incident caused by biological processes not covered by specific subcategories. Examples: a fungal contamination of a water source, an algal bloom affecting a reservoir.
49	Climate.Drought	A prolonged period of below-average precipitation leading to a water shortage, which can impact agriculture, ecosystems, and water supplies. Examples: a multi-year drought affecting a region, a dry spell causing crop failure.
50	Climate.LakeOutburst	An incident where a glacial lake dam (often moraine or ice) fails, rapidly releasing a large volume of water and causing devastating floods downstream. Examples: a glacial lake outburst flood in the Alps, a moraine dam breach in the Himalayas.
51	Climate.Wildfire	An unplanned and uncontrolled fire burning in natural or rural areas, often exacerbated by climatic conditions like drought, wind, and heat. Examples: a wildfire spreading under heat and wind, a forest fire threatening a settlement.
52	Climate.Other	Any other incident caused by long-lived atmospheric processes (climatological) not covered by specific subcategories. Examples: a prolonged heat dome, a large-scale shift in monsoon patterns.

Rank	Keyword	Description
53	Extraterrestrial.Aliens	A hypothetical incident involving the discovery of or interaction with extraterrestrial intelligent life, a theoretical scenario in scientific and security planning. Examples: a hypothetical first-contact scenario, a tabletop exercise on extraterrestrial contact.
54	Extraterrestrial.Impact	An incident involving a celestial object (asteroid, meteoroid, comet) colliding with Earth, potentially causing localized or global damage. Examples: a meteor airburst over a populated area, an asteroid impact warning.
55	Extraterrestrial.SpaceWeather	An event caused by solar or cosmic activity, such as solar flares or geomagnetic storms, that can disrupt Earth's technological infrastructure. Examples: a geomagnetic storm disrupting power grids, satellite link degradation during a solar flare.
56	Extraterrestrial.Other	Any other incident caused by extraterrestrial processes not covered by specific subcategories. Examples: a cosmic ray burst affecting electronics, an unusual solar particle event.
57	Fraud.Copyright	The act of reproducing, distributing, or installing software, media, or other materials in violation of their copyright, often for personal gain or distribution (piracy). Examples: illegal distribution of licensed software, the sale of pirated media.
58	Fraud.Corruption	A fraudulent scheme that is made possible by the abuse of power or position by a trusted individual (e.g., employee, official) who acts for personal gain. Examples: an admin granting rights for a bribe, an official approving a fraudulent permit.
59	Fraud.Espionage	The use of illegal or unethical means, such as hacking,

Rank	Keyword	Description
		bribery, or theft, to acquire a competitor's trade secrets, intellectual property, or other confidential business information. Examples: exfiltration of trade secrets, photographing a prototype.
60	Fraud.Masquerade	A type of attack where an attacker illegitimately assumes the identity of another user, process, or system to gain unauthorized access, privileges, or benefits. Examples: an attacker impersonating a service account, a person using someone else's badge.
61	Fraud.Usage	The use of an organization's assets (e.g., computing power, network, email) for non-work-related, often illegal, activities without authorization. Examples: mining crypto on corporate servers, personal use of a company vehicle.
62	Fraud.Other	Any other incident involving deception for financial or reputational gain that does not fit into the specific fraud subcategories. Examples: a fake refund scheme, a forged signature on a contract.
63	Geophysical.Earthquake	An incident caused by a sudden, rapid shaking of the earth resulting from the movement of tectonic plates, which can cause ground shaking, surface rupture, and tsunamis. Examples: a magnitude 6.0 earthquake, an aftershock swarm following a main shock.
64	Geophysical.MassMovement	An incident involving the downslope movement of rock, soil, or snow under the force of gravity, such as landslides, avalanches, or rockfalls. Examples: a landslide after saturation, an avalanche blocking a mountain road.
65	Geophysical.Other	Any other incident caused by solid-earth processes not covered by specific geophysical subcategories. Examples:

Rank	Keyword	Description
		a volcanic tremor, ground deformation detected by sensors.
66	Geophysical.Volcanic	An incident caused by the eruption of magma and volcanic gases from a volcano, which can produce lava flows, pyroclastic flows, ashfall, and lahars. Examples: ashfall from an eruption, a pyroclastic flow descending a slope.
67	Hydro.Flood	An incident where water submerges land that is normally dry, often caused by heavy rain, storm surge, or dam failure, leading to property damage and risk to life. Examples: a river flood after heavy rain, a flash flood in an urban area.
68	Hydro.Landslide	An incident involving the downward movement of slope materials (soil, rock) triggered by water saturation from heavy rain or snowmelt. Examples: a rain-triggered slope failure, a mudslide after snowmelt.
69	Hydro.Wave	An incident involving destructive waves, such as tsunamis or storm surges, that can cause coastal flooding, erosion, and damage. Examples: a tsunami hitting a coast, a storm surge flooding a seafront.
70	Hydro.Other	Any other incident caused by the movement, distribution, and quality of water, not covered by specific hydrological subcategories. Examples: a dam overflow warning, a sudden drop in river water quality.
71	Insider.Malicious	A security incident caused by a current or former employee, contractor, or other trusted insider who intentionally acts to harm the organization, its data, or its people. Examples: an employee exfiltrating data, an insider disabling a camera.
72	Insider.Negligent	A security incident caused unintentionally by an insider, such as through carelessness, lack of awareness, or simple

Rank	Keyword	Description
		human error, leading to data exposure or system compromise. Examples: a mis-sent email with PII, leaving a door unlocked.
73	Insider.Other	Any other security incident involving an insider (trusted individual) that does not fit into the specific insider threat subcategories. Examples: a contractor using shadow IT, a visitor wandering off from an escorted area.
74	Malware.Adware	A type of software that automatically displays or downloads unwanted advertisements, often in a disruptive manner, and may track user behavior. Examples: a browser hijacker showing ads, unwanted pop-ups appearing on a workstation.
75	Malware.Backdoor	A type of malware that bypasses normal authentication to give an attacker persistent remote access to a compromised system, often enabling data theft or installation of additional malware. Examples: a RAT installed on a workstation, a hidden remote shell on a server.
76	Malware.ResourceHijacking	A type of malware or malicious behavior that abuses the resources of a compromised system (CPU, GPU, network bandwidth, cloud services, etc.) for the attacker's benefit without the owner's consent. This includes, but is not limited to, cryptocurrency mining (cryptojacking), proxy bandwidth resale, SMS traffic generation, and email spam abuse. The main impact is degraded performance, reduced availability, and increased operational costs (electricity, cloud billing). Examples: XMRig running on a server, a browser-based miner slowing a workstation.

Rank	Keyword	Description
77	Malware.Downloader	A type of malware designed to download and install other malicious software from a remote server, typically acting as the first stage in a multi-step attack. Examples: a dropper fetching a payload, a script downloading a second-stage binary.
78	Malware.Ransomware	A type of malware that encrypts a victim's files or systems, rendering them inaccessible, and demands a ransom payment, often in cryptocurrency, for the decryption key. Examples: file-encrypting malware on a file server, a ransom note left on a workstation.
79	Malware.Rootkit	A type of malware designed to hide its presence and grant an attacker persistent, privileged access to a compromised system while evading detection. Examples: a kernel-level rootkit, a bootkit persisting across reboots.
80	Malware.Spyware	A type of software that secretly monitors and collects information about a user's activities, such as keystrokes, browsing habits, and personal data, without their consent. Examples: a keylogger on a laptop, a hidden camera implant.
81	Malware.Trojan	A type of malware that disguises itself as legitimate or desirable software to trick users into installing it, after which it can perform malicious actions. Examples: a fake installer, a trojanised game download.
82	Malware.Virus	A type of malicious software that attaches itself to a legitimate program or file and replicates itself to spread to other programs, often requiring human action to propagate. Examples: a macro virus in a document, an infected executable on a USB stick.

Rank	Keyword	Description
83	Malware.Worm	A type of standalone malicious software that replicates itself to spread across networks, often exploiting vulnerabilities without requiring human interaction. Examples: a self-spreading worm on a network, a mass-mailing worm.
84	Malware.Other	Any other incident involving malicious software that does not fit into the specific malware subcategories. Examples: fileless malware, a malicious browser extension.
85	Meteo.Cold	A meteorological incident involving a rapid fall in temperature over a short period, or a prolonged period of extreme cold, posing risks to health and infrastructure. Examples: a cold wave, a sudden frost damaging crops.
86	Meteo.Fog	A meteorological incident where dense fog reduces visibility, potentially disrupting transportation and causing accidents. Examples: dense fog on a motorway, fog grounding flights at an airport.
87	Meteo.Heat	A meteorological incident involving a prolonged period of excessively hot weather, which can cause health impacts, infrastructure stress, and environmental damage. Examples: a heatwave, a prolonged period of high night-time temperatures.
88	Meteo.Rain	A meteorological incident involving excessive or prolonged rainfall that can lead to flooding, landslides, and transportation disruptions. Examples: prolonged rainfall over a catchment, intense rainfall causing urban flooding.
89	Meteo.Snow	A meteorological incident involving heavy snowfall and blizzard conditions, which can disrupt transportation, damage infrastructure, and pose risks to safety. Examples: a blizzard,

Rank	Keyword	Description
		heavy snow closing a mountain pass.
90	Meteo.Wind	A meteorological incident involving damaging or dangerous winds, such as from storms, tornadoes, or hurricanes, that can cause structural damage and power outages. Examples: storm-force winds, a tornado touching down.
91	Meteo.Other	Any other incident caused by atmospheric processes not covered by specific meteorological subcategories. Examples: a hail storm, a dust storm reducing visibility.
92	National.Conflict	A state of armed conflict between two or more nations, involving organized military forces and impacting national security. Examples: an armed border clash, a state-backed cyber campaign during a war.
93	National.Crime	Large-scale illegal activities, such as trafficking in drugs, weapons, or people, conducted by sophisticated criminal networks that pose a threat to national and international security. Examples: a ransomware cartel, drug trafficking across a border.
94	National.Cyber	Large-scale cyber operations, including attacks, espionage, and disinformation campaigns, conducted by state-sponsored or state-affiliated groups against national interests. Examples: a state-sponsored APT campaign, a coordinated disinformation operation.
95	National.Economic	An incident involving significant disruption to a nation's economy, such as market crashes, hyperinflation, or trade wars, affecting national stability. Examples: a market crash, a sudden currency devaluation.
96	National.Environmental	Incidents related to large-scale environmental shifts, such as climate change,

Rank	Keyword	Description
		resource scarcity, or global pandemics, that have significant national security implications. Examples: climate-driven migration, a resource scarcity crisis.
97	National.Societal	An incident involving large-scale public disorder, such as strikes, riots, or civil unrest, that challenges social order and may require national-level response. Examples: a general strike, widespread riots in a city.
98	National.Terrorism	An incident involving the use of violence, intimidation, or threats by non-state actors against civilians or property to achieve political, religious, or ideological objectives. Examples: a bombing in a public place, online radicalization leading to a plot.
99	National.Other	Any other incident that has a significant impact at the national scale, not covered by specific national security subcategories. Examples: a national emergency declaration, a large-scale border incident.
100	Operational.Configuration	A change, whether intentional or accidental, to the parameters, settings, or behavior of a system, service, or component. Examples: a firewall rule change, switching a valve in a pipeline.
101	Operational.Misuse	The use of an organization's assets (e.g., computers, network, time) for purposes that are not officially authorized, which may violate policy but not be malicious. Examples: personal use of a corporate account, personal use of a company vehicle.
102	Operational.PolicyViolation	An incident where an individual fails to comply with established organizational policies, procedures, or security rules, whether intentionally or accidentally. Examples:

Rank	Keyword	Description
		bypassing a security policy, ignoring a safety rule.
103	Operational.ProcessFailure	An incident where a designed process, procedure, or workflow fails to achieve its intended outcome, leading to operational or security gaps. Examples: a failed patch rollout, a broken maintenance procedure.
104	Operational.Update	The installation, upgrade, or refresh of software, firmware, signatures, definitions, or other content that modifies the functional behavior of a system, service, or component. Examples: a software upgrade, a firmware update on a device.
105	Operational.Other	Any other operational issue that does not fit into the specific operational incident subcategories. Examples: a license expiry, a spare part shortage.
106	Recon.Aerial	The act of using aerial platforms, such as drones or satellites, to conduct surveillance or gather intelligence about a target area. Examples: a drone overflight, satellite imagery collection over a site.
107	Recon.Landscape	The act of physically observing a location to gather information about security measures, layouts, or potential vulnerabilities. Examples: casing a building, observing a facility's entry points.
108	Recon.Network	The practice of probing a network to discover active hosts, open ports, and services, often as a precursor to an attack or unauthorized monitoring. Examples: a port scan, a host discovery sweep.
109	Recon.OSINT	The practice of collecting and analyzing information from publicly available sources (e.g., social media, public records) for intelligence purposes. Examples: scraping social

Rank	Keyword	Description
		media, gathering information from public records.
110	Recon.Other	Any other incident involving scanning, spying, or monitoring to identify resources that does not fit into specific recon subcategories. Examples: DNS enumeration, monitoring of a target's communications.
111	Sabotage.Data	The intentional act of deleting, altering, or corrupting digital or physical data to cause harm, disrupt operations, or cover tracks. Examples: a database wiped, shredded documents.
112	Sabotage.Destruction	The intentional and malicious act of destroying or severely damaging physical assets, such as windows, equipment, or buildings. Examples: smashed equipment, a deliberately damaged vehicle.
113	Sabotage.Disruption	The intentional disruption of essential services or utilities, such as electricity, water, or network connectivity, to cause operational downtime. Examples: a cut fibre link, a cut power line.
114	Sabotage.Equipment	The intentional act of causing damage to operational equipment, machinery, or vehicles, often to disrupt production or operations. Examples: a damaged machine on a production line, a sabotaged vehicle.
115	Sabotage.Graffiti	The act of willfully defacing, damaging, or marking public or private property with inscriptions, drawings, or tags without permission. Examples: tags on a wall, graffiti on a train.
116	Sabotage.Tampering	The act of deliberately meddling with or disabling security controls (e.g., locks, alarms, cameras) to compromise their effectiveness. Examples: a disabled EDR agent, a disabled camera.
117	Sabotage.Vandalism	The deliberate and malicious act of damaging, destroying, or

Rank	Keyword	Description
		obstructing an organization's physical assets, operations, or systems. Examples: broken windows, damaged signage.
118	Sabotage.Other	Any other incident involving the intentional damage to property or assets not covered by the specific sabotage subcategories. Examples: glue in a lock, a blocked ventilation shaft.
119	Safety.Accident	An unplanned, unforeseen event (e.g., vehicle crash, industrial mishap, chemical spill) that results in injury, loss of life, or damage to health. Examples: a vehicle crash, a fall from height.
120	Safety.Aggression	An incident where an individual uses physical force against another person, causing bodily harm, pain, or the fear of immediate harm. Examples: a physical assault, a violent altercation.
121	Safety.Explosion	A sudden, violent release of energy (e.g., from gas, chemicals, or explosives) that causes a blast, fire, and potential structural damage, injury, or loss of life. Examples: a gas explosion, a chemical blast.
122	Safety.Fire	An incident involving uncontrolled burning (e.g., structural, wildland, or chemical fire) that threatens human safety, property, or the environment. Examples: a data-center fire, a building fire, a wildfire.
123	Safety.Hostage	An incident where a person or group is held against their will by a captor, often to compel a third party to meet certain demands. Examples: a hostage-taking in a building, a person held in a vehicle.
124	Safety.Pollution	The introduction or abnormal presence of a contaminant (chemical, biological, radioactive, or otherwise hazardous) in the environment

Rank	Keyword	Description
		— air, water, soil, or built infrastructure — that causes or may cause harm to human health, ecosystems, property, or the operation of facilities and services. The pollution may be acute or chronic, accidental or deliberate, and of local or transboundary extent. Examples: a radioactive release from a nuclear facility, a chemical discharge into a river, an atmospheric pollution peak, soil contamination from an industrial site, a deliberate dumping of hazardous waste.
125	Safety.Sexual	An incident involving any unwanted sexual act, contact, or behavior directed against an individual without their consent. Examples: a sexual assault, unwanted sexual contact.
126	Safety.Other	Any other incident that causes or has the potential to cause injury, loss of life, or endanger citizens, not covered by specific safety subcategories. Examples: a fall from height, a hazardous material spill.
127	SocialEng.Baiting	A social engineering attack that lures victims by offering something desirable (e.g., free music, a prize) in exchange for information or by tricking them into downloading malware. Examples: a USB drop in a car park, a left USB in a lobby.
128	SocialEng.Phishing	A social engineering attack where the attacker impersonates a trustworthy entity to trick a victim into revealing sensitive information or performing an action. Examples: a credential-harvesting email, a fake invoice sent by post, a fake login page.
129	SocialEng.Pretexting	A social engineering tactic where the attacker creates a fabricated scenario or pretends to be someone they are not to engage a victim and extract information or access.

Rank	Keyword	Description
		Examples: a fake IT support call, a fake inspector visit.
130	SocialEng.QuidProQuo	A social engineering tactic where the attacker offers a service or benefit (e.g., technical support) in exchange for information or access, often expecting something in return later. Examples: a fake software update, a bogus tech support offer.
131	SocialEng.Other	Any other incident involving psychological manipulation of people to divulge information or perform actions, not covered by specific social engineering subcategories. Examples: a deepfake voice call, a fake job interview used to extract information.
132	SupplyChain.Compromise	A security incident where an attacker exploits a vulnerability in a third-party vendor's system to gain access to or compromise the primary target's network or data. Examples: a compromised software update, a tampered shipment.
133	SupplyChain.Disruption	An event that disrupts the normal flow of products, services, or information within a supply chain, often impacting operations and delivery. Examples: a cloud provider outage, a blocked port.
134	SupplyChain.Other	Any other incident affecting the supply chain that does not fit into the specific supply chain subcategories. Examples: an expired certificate from a vendor, delayed spare parts.
135	Theft.Data	The unauthorized taking or copying of sensitive or confidential documents, whether in physical or digital form. Examples: copied files from a shared drive, stolen paper records.
136	Theft.Equipment	The unlawful taking of physical hardware, such as computers, mobile phones, or servers, resulting in loss of assets and potentially the data they

Rank	Keyword	Description
		contain. Examples: a stolen server, a stolen laptop.
137	Theft.FinInfo	The theft of financial information, such as credit card numbers or bank account details, often for the purpose of fraudulent transactions. Examples: card skimming data, a stolen wallet.
138	Theft.IP	The unlawful acquisition of a company's intellectual property, including trade secrets, patents, formulas, or proprietary processes. Examples: stolen source code, a stolen blueprint.
139	Theft.Machinery	The unlawful taking of heavy equipment, vehicles, or specialized machinery, often resulting in significant operational and financial loss. Examples: a stolen excavator, a stolen delivery truck.
140	Theft.PII	The unauthorized acquisition of Personally Identifiable Information (PII) that can be used to identify, contact, or impersonate an individual. Examples: a leaked customer database, stolen HR files.
141	Theft.Other	Any other incident involving the intentional stealing of physical property or digital data not covered by specific theft subcategories. Examples: a stolen license key, stolen tools.
142	Other.Uncategorized	Any incident that does not fit into one of the predefined categories in this taxonomy. Examples: an unclassifiable alert, an incident whose type does not match any existing category.
143	Other.Undetermined	An incident whose category is currently unknown, under investigation, or cannot be determined. Examples: an alert under investigation, a signal whose cause remains unidentified.
144	Other.Test	An incident generated solely for the purpose of testing systems, processes, or training personnel.

Rank	Keyword	Description
		Examples: a synthetic test alert, a fire drill.
145	Other.ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1) Examples: an extension placeholder used when no standard category applies.

Table 10: enum_alert_category

C.3. Analyzer Category

uank	Keyword	Description
0	Undetermined	The analyzer category has not been determined, or cannot be established from available information. Ex: an analyzer whose type is unknown at ingestion time.
1	APP.BAST	Bastion Host — A hardened, intermediary server that provides secure remote access to a protected network, acting as a single controlled entry point. Ex: an SSH jump host, a RDP gateway.
2	APP.DAST	Dynamic Application Security Testing — Testing a running application from the outside to find security weaknesses in its runtime behaviour, without access to source code. Ex: a scanner probing a live web app for injection flaws.
3	APP.IAST	Interactive Application Security Testing — Hybrid testing that instruments a running application from within (agent-based) while it is exercised, combining runtime and code-level visibility. Ex: an agent inside a Java app reporting tainted data flows.
4	APP.RASP	Runtime Application Self-Protection — Security embedded inside a running application that detects and blocks attacks in real time from within the runtime, independently of the

uank	Keyword	Description
		surrounding infrastructure. Ex: an in-app module blocking a SQL injection attempt.
5	APP.SANDBOX	Sandbox / File Detonation — Isolated execution environment that analyses the behaviour of suspicious files or URLs and decides whether to raise an alert. Ex: a suspicious attachment detonated in a VM to observe its behaviour.
6	APP.SAST	Static Application Security Testing — Analysis of application source code or binaries without executing them, to find vulnerabilities early in development. Ex: a linter flagging unsanitised input in source.
7	APP.SCA	Software Composition Analysis — Analysis of third-party dependencies and libraries for known vulnerabilities, licence issues, and supply-chain risk. Ex: detecting a vulnerable open-source component in a build.
8	CLD.CASB	Cloud Access Security Broker — A control point between users and cloud services that enforces security policy, visibility, and compliance for cloud usage. Ex: blocking unsanctioned SaaS uploads.
9	CLD.CIEM	Cloud Infrastructure Entitlement Management — Management and least-privilege enforcement of identities, roles, and permissions across cloud environments. Ex: detecting over-permissive IAM roles.
10	CLD.CSPM	Cloud Security Posture Management — Continuous monitoring of cloud configurations for misconfigurations, drift, and compliance violations. Ex: flagging a publicly exposed storage bucket.
11	CLD.CWPP	Cloud Workload Protection Platform — Security for cloud workloads (VMs, containers, serverless) covering runtime,

uank	Keyword	Description
		vulnerability, and threat detection. Ex: detecting a cryptominer in a container.
12	DDoS.ANTI-DDOS	Distributed Denial of Service Protection — Systems and techniques that detect and mitigate volumetric or application-layer DDoS attacks. Ex: rate-limiting and scrubbing malicious floods.
13	DDoS.SCRUB	Scrubber / Scrubbing Center — A dedicated traffic-cleaning facility that diverts and filters attack traffic before forwarding clean traffic to the target. Ex: a carrier scrubbing centre absorbing a 1 Tbps flood.
14	DDoS.WAF-DDOS	Web Application Firewall with DDoS Protection — A WAF that also provides DDoS mitigation for HTTP/HTTPS traffic. Ex: blocking a Layer 7 HTTP flood against a web service.
15	EMAIL.ANTI-PHISH	Anti-Phishing — Detection and blocking of phishing attempts in email, including malicious links, spoofed senders, and deceptive content. Ex: quarantining a credential-harvesting email.
16	EMAIL.DMARC	Domain-based Message Authentication, Reporting and Conformance — Email authentication (SPF/DKIM/ DMARC) monitoring and enforcement to prevent sender spoofing. Ex: rejecting mail failing DMARC alignment.
17	EMAIL.SEG	Secure Email Gateway — A comprehensive email security appliance/service filtering inbound and outbound mail for spam, malware, phishing, and policy violations. Ex: a gateway scanning attachments before delivery.
18	EMAIL.SPAM-FILTER	Spam Filter — Detection and filtering of unsolicited bulk email based on content, reputation, and heuristics. Ex: moving a bulk advertisement to the junk folder.

uank	Keyword	Description
19	END.AM	Application Allowlisting — Execution control that permits only explicitly approved applications to run, blocking all others by default. Ex: a workstation that only runs signed corporate apps.
20	END.AV	Antivirus — Signature- and heuristic-based detection and removal of known malware on endpoints. Ex: quarantining a file matching a known virus signature.
21	END.DLP-EP	Endpoint Data Loss Prevention — Prevention of sensitive data leakage from endpoint devices via channels such as USB, clipboard, print, or email. Ex: blocking a copy of customer data to a USB stick.
22	END.EDR	Endpoint Detection and Response — Continuous endpoint monitoring, behavioural detection, and response capabilities for advanced threats. Ex: isolating a host showing lateral-movement behaviour.
23	END.EPP	Endpoint Protection Platform — A consolidated endpoint security suite combining antivirus, firewall, EDR, and other protections in a single agent. Ex: a single agent providing AV, host firewall, and threat hunting.
24	END.HIDS	Host Intrusion Detection System — Detection of malicious or anomalous activity on a host by monitoring system calls, files, and logs. Ex: alerting on a suspicious process spawning from a web server.
25	END.HIPS	Host Intrusion Prevention System — Active prevention of host-level intrusions by blocking malicious system activity in real time. Ex: blocking an unauthorised process from modifying system files.
26	END.HPT	Honeypot — A decoy system or service designed to attract and

uank	Keyword	Description
		detect attackers, providing early warning and intelligence. Ex: a fake SSH server logging brute-force attempts.
27	HUM.REPORT	Human Report — A human observer who analyses a situation and decides to raise an alert, acting as the analyzer in the IDMEFv2 sense. Ex: a guard reporting a suspicious crowd approaching a building, an operator qualifying an incident in a control room, a patrol officer signalling an anomaly.
28	ID.DCAP	Data-Centric Audit and Protection — Monitoring and auditing of data access and usage, focused on the data itself rather than the infrastructure. Ex: alerting on unusual access to a sensitive database.
29	ID.DLP	Data Loss Prevention — Detection and prevention of sensitive data exfiltration across all channels (network, endpoint, cloud). Ex: blocking an outbound email containing PII.
30	ID.IAM	Identity and Access Management — Governance of identities and enforcement of access control across systems and applications. Ex: provisioning, authenticating, and authorising user accounts.
31	ID.IRM	Identity Risk Management — Assessment and scoring of identity-related risk, such as dormant accounts, excessive privileges, or compromised credentials. Ex: flagging a service account with no owner.
32	ID.MFA	Multi-Factor Authentication Analyzer — Analysis of multi-factor authentication events to detect bypass attempts, repeated failures, or anomalous challenges. Ex: alerting on repeated MFA push notifications ignored by a user.
33	ID.PAM	Privileged Access Management — Control, vaulting, and

uank	Keyword	Description
		monitoring of privileged accounts and sessions. Ex: a vault checking out an admin password for a session.
34	ID.PIM	Privileged Identity Management — Just-in-time and lifecycle management of privileged identities and roles. Ex: granting temporary admin rights for a maintenance window.
35	ID.SECRETS	Secrets Detection — Analysis of code repositories, configurations, and logs to detect exposed secrets such as API keys, tokens, or certificates. Ex: detecting an API key committed to a public repository.
36	ID.UEBA	User and Entity Behaviour Analytics — Behavioural analytics over users and entities to detect anomalies and insider threats. Ex: alerting on a user downloading an unusual volume of files.
37	NAT.AIR	Air Quality Sensor — Analysis of airborne particles and gases to detect pollution, smoke, or hazardous releases. Ex: detecting a pollution peak or a smoke plume from a fire.
38	NAT.FIRE	Wildfire Detection — Detection of uncontrolled fires in natural or rural areas, typically via thermal or infrared sensing. Ex: a thermal sensor detecting a wildfire starting in a forest.
39	NAT.HYDRO	Hydrological Sensor — Analysis of water levels, flow, and quality to detect floods, droughts, or contamination. Ex: a river gauge alerting on a rising water level.
40	NAT.METEO	Meteorological Sensor — Analysis of atmospheric conditions (temperature, wind, precipitation, pressure) to detect weather events. Ex: a weather station alerting on a heatwave or storm.
41	NAT.RADIO	Radiation Sensor — Analysis of ionizing radiation levels

uank	Keyword	Description
		to detect nuclear incidents or radioactive releases. Ex: a radiation monitor alerting on abnormal levels near a facility.
42	NAT.SAT	Earth Observation Satellite — Analysis of satellite imagery to detect large-scale natural events, environmental changes, or disasters. Ex: satellite imagery detecting a cyclone, a wildfire, or a flood.
43	NAT.SEISMO	Seismic Sensor — Analysis of ground motion to detect earthquakes, volcanic activity, or subsurface events. Ex: a seismograph alerting on a magnitude 6.0 earthquake.
44	NET.DNS-FW	DNS Firewall — Filtering of DNS queries to block resolution of malicious or unwanted domains. Ex: blocking resolution of a known C2 domain.
45	NET.DPI	Deep Packet Inspection — Inspection of packet payloads beyond headers to identify protocols, applications, or malicious content. Ex: detecting malware signatures inside HTTPS traffic.
46	NET.FW	Firewall — Filtering and policy enforcement of network traffic based on addresses, ports, and protocols. Ex: blocking inbound traffic to a database port from the internet.
47	NET.NAC	Network Access Control — Enforcement of endpoint compliance and identity before granting network access. Ex: quarantining a non-patched laptop until it is remediated.
48	NET.NBAD	Network Behaviour Anomaly Detection — Detection of deviations from normal network behaviour patterns, typically via baselining and statistics. Ex: alerting on an unusual spike in DNS queries from one host.
49	NET.NDR	Network Detection and Response — Continuous network monitoring, behavioural detection, and

uank	Keyword	Description
		response for advanced threats. Ex: correlating east-west traffic to identify lateral movement.
50	NET.NGFW	Next-Generation Firewall — A firewall with application awareness, user identity, and integrated threat prevention (IPS, AV, URL filtering). Ex: allowing only sanctioned apps regardless of port.
51	NET.NIDS	Network Intrusion Detection System — Passive monitoring of network traffic to detect known attack patterns and anomalies. Ex: alerting on a signature matching an exploit attempt.
52	NET.NIPS	Network Intrusion Prevention System — Inline inspection of network traffic that can actively block detected attacks. Ex: dropping packets matching an exploit signature.
53	NET.PROXY	Proxy Server — An intermediary for client requests that enforces ACLs and can inspect TLS sessions. Ex: a forward proxy logging and filtering outbound web traffic.
54	NET.WAF	Web Application Firewall — Filtering of HTTP/ HTTPS traffic to protect web applications from common attacks (OWASP Top 10, etc.). Ex: blocking a SQL injection attempt against a web app.
55	NET.WIDS	Wireless Intrusion Detection System — Passive monitoring of wireless networks to detect rogue APs, evil twins, and wireless attacks. Ex: alerting on an unauthorised AP in the building.
56	NET.WIPS	Wireless Intrusion Prevention System — Active prevention of wireless threats, including containment of rogue devices. Ex: deauthenticating clients from a rogue AP.
57	OT.IOT-IDS	IoT Intrusion Detection System — Detection of threats targeting IoT devices, whose constraints (power, compute,

uank	Keyword	Description
		protocols) differ from general IT. Ex: alerting on anomalous MQTT traffic from a sensor.
58	OT.OT-IDS	Operational Technology IDS — Detection of threats in industrial control systems (SCADA, DCS), where availability and safety dominate. Ex: alerting on unauthorised Modbus writes to a PLC.
59	OT.PLC-SCAN	PLC Scanner — Discovery and vulnerability assessment of PLC/SCADA devices in operational environments. Ex: enumerating PLCs on a plant network and checking firmware versions.
60	OT.SCADA	SCADA Monitor — Analysis of supervisory control and data acquisition systems to detect abnormal process states or operational anomalies. Ex: alerting on an abnormal pressure reading in an industrial process.
61	OT.SENSOR	Industrial Sensor — Analysis of industrial measurements (pressure, temperature, flow) to detect abnormal process conditions. Ex: a flow sensor alerting on a sudden drop in a pipeline.
62	PHY.1D-LAS	Laser Sensor 1D — Measures distance or presence along a single axis using a laser beam. Ex: a laser rangefinder, a beam-break tripwire.
63	PHY.2D-LAS	Laser Sensor 2D — Scans a plane with a laser to detect presence, contours, or movement in two dimensions. Ex: a planar laser scanner guarding an entrance.
64	PHY.3D-LAS	Laser Sensor 3D — Scans a volume with a laser to produce a three-dimensional representation of a scene. Ex: a 3D scanner producing a point cloud of a room.
65	PHY.ACCESS-CTRL	Access Control System — Physical entry/exit control using credentials (badge, PIN,

uank	Keyword	Description
		biometric) and monitoring of door state. Ex: a badge reader logging who enters a server room.
66	PHY.ADS	Anti-Drone System — Detection (and sometimes neutralisation) of unauthorised drones via RF, radar, or acoustic sensing. Ex: detecting a drone approaching a restricted site.
67	PHY.CBRN	CBRN Detector — Detection of chemical, biological, radiological, or nuclear agents in the environment. Ex: a detector alerting on a chemical agent release.
68	PHY.FIRE	Fire Detector — Detection of heat, flame, or combustion products indicating a fire. Ex: a flame detector triggering an alarm in a warehouse.
69	PHY.FR-CAM	Face Recognition Camera — A camera coupled with biometric software that identifies or verifies individuals from facial features. Ex: matching a person against an authorised list at a gate.
70	PHY.GAS	Gas Detector — Detection of hazardous or combustible gases in the environment. Ex: a methane detector alerting in a confined space.
71	PHY.GLASS-BRK	Glass Break Detector — An acoustic sensor detecting the specific sound signature of breaking glass. Ex: triggering an alarm when a window is smashed.
72	PHY.HAR	Human Activity Recognition — AI-based analysis of video or sensor data to recognise human activities and behaviours. Ex: detecting a person falling, running, or loitering.
73	PHY.LWIR	Long-Wave Infrared — Thermal imaging in the long-wave IR band (roughly 8–14 µm), suited to ambient-temperature detection. Ex: a thermal camera detecting body heat at night.

uank	Keyword	Description
74	PHY.MOT-SEN	Motion Sensor — A sensor (PIR, microwave, ultrasonic) detecting movement in its field of view. Ex: a PIR sensor turning on a light and logging presence.
75	PHY.MWIR	Mid-Wave Infrared — Thermal imaging in the mid-wave IR band (roughly 3–5 µm), used for higher-temperature targets and some gas detection. Ex: a MWIR camera monitoring an industrial furnace.
76	PHY.OBJ-DET	Object Detection Camera — A camera with AI models that detect and classify objects in the scene. Ex: detecting a left bag or a vehicle in a restricted zone.
77	PHY.PERIMETER	Perimeter Detection — Detection of intrusion across a perimeter using barriers, fibre optics, or microwave sensing. Ex: a fence-mounted fibre sensor alerting on an attempted climb.
78	PHY.RADAR	Radar Sensor — Detection of movement, presence, or objects using radio waves. Ex: a ground radar detecting an intrusion across a perimeter.
79	PHY.SMOKE	Smoke Detector — Detection of smoke particles indicating a fire. Ex: a smoke detector triggering an alarm in a server room.
80	PHY.SWIR	Short-Wave Infrared — Imaging in the short-wave IR band (roughly 0.9–1.7 µm), useful for seeing through haze and some materials. Ex: a SWIR camera inspecting contents through a container.
81	PHY.VAD	Voice Activity Detection — Audio analysis that detects the presence of human speech in a signal. Ex: triggering recording only when speech is present.
82	PHY.VNIR	Visible and Near-Infrared — Multi-spectral imaging covering the visible and near-IR bands. Ex: a VNIR camera used in low-light surveillance.

uank	Keyword	Description
83	PHY.WATER	Water Leak Detector — Detection of water presence or leakage in a monitored area. Ex: a leak sensor alerting in a data centre.
84	SIEM.APM	Application Performance Monitoring — Analysis of application performance and availability to detect degradations, errors, or outages. Ex: alerting on a spike in response time for a critical service.
85	SIEM.ETL	Extract, Transform, Load — Data pipeline tooling that collects, normalises, and routes data to downstream stores. Ex: Logstash, Fluentd, Vector.
86	SIEM.LOG	Log Analyzer — Aggregation, indexing, and search of log data for troubleshooting and security analysis. Ex: an ELK stack or Splunk index.
87	SIEM.NMS	Network Management System — Monitoring and management of network devices, links, and performance. Ex: a NMS alerting on a down interface.
88	SIEM.SIEM	Security Information and Event Management — Centralised collection, correlation, and alerting across security-relevant data sources. Ex: correlating firewall and endpoint events into one alert.
89	SIEM.SOAR	Security Orchestration, Automation and Response — Automation of security workflows and incident response playbooks. Ex: auto-enriching an alert and opening a ticket.
90	SIEM.UPTIME	Uptime Monitoring — Analysis of service and infrastructure availability to detect outages and recoveries. Ex: alerting when a critical service becomes unreachable.
91	TI.CTI	Cyber Threat Intelligence — Strategic and operational intelligence about threats, actors, and campaigns, produced for decision-making.

uank	Keyword	Description
		Ex: a report on an APT targeting your sector.
92	TI.TI-FEED	Threat Intelligence Feed — A stream of structured threat data (IOCs, TTPs) consumed for detection and enrichment. Ex: a STIX/TAXII feed of malicious IPs.
93	TI.TIP	Threat Intelligence Platform — Aggregation, correlation, and management of threat intelligence from multiple sources. Ex: a platform correlating IOCs from several feeds.
94	VM.ASM	Attack Surface Management — Discovery and continuous monitoring of an organisation's externally exposed assets. Ex: finding an unknown public subdomain.
95	VM.PENTEST	Penetration Testing Tools — Tools supporting manual or automated offensive testing of systems and applications. Ex: exploitation frameworks and scanners used by testers.
96	VM.VULN-SCANNER	Vulnerability Scanner — Automated assessment of systems and applications for known vulnerabilities. Ex: a scanner reporting missing patches and CVEs.
97	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1) Ex: a vendor-specific analyzer type declared through the extension mechanism.

Table 11: enum_analyzer_category

C.4. Analyzer Data

Rank	Keyword	Description
0	Undetermined	The analyzed data type is unknown or not yet classified. Used as a placeholder when no other value applies. Example: a generic analyzer whose input could not be identified.

Rank	Keyword	Description
1	Acoustics	Sound pressure or specific frequency measurements captured by a microphone or acoustic sensor. Example: gunshot detection, glass break detection, machine noise spectrum.
2	Alert	IDMEFv2 or equivalent alert produced by an upstream analyzer and consumed as input by a correlator, aggregator or hypervisor. Example: IDMEFv2 JSON alert from a local SOC processed by a central SOC correlator.
3	Contact	Physical interaction or binary state change detected by a contact sensor. Example: door contact switch, cabinet open detection, tamper switch.
4	Content	Payload or body of a communication inspected at the application layer. Example: HTTP body, email message, SMTP DATA section, file transfer payload, DPI-matched bytes.
5	Data	Generic or opaque data stream whose structure is not further specified. Example: raw sensor telemetry, vendor-specific binary stream, serial output.
6	Datagram	Raw packet-level data captured at the network layer. Example: Ethernet frame, IPv4/IPv6 packet, TCP segment, UDP datagram, ICMP message.
7	External	Data originating from a third party such as a threat intelligence feed or an external CERT. Example: STIX/TAXII feed, IOC list, external CERT notification, vendor advisory.
8	File	Content or metadata of a file read by the analyzer for inspection, integrity or change detection. Example: file hash, PE header, document content, file permissions, modification timestamp.
9	Flow	Network flow records aggregating packets by 5-tuple or similar criteria. Example:

Rank	Keyword	Description
		NetFlow, IPFIX, sFlow records exported by routers or probes.
10	Fog	Visibility reduction or optical density measured by a fog sensor. Example: fog detection on a perimeter road, visibility monitoring for cameras.
11	Humidity	Relative humidity or moisture content in the air measured by a hygrometer. Example: datacenter humidity level, warehouse moisture monitoring.
12	Images	Visible-spectrum image or video frames captured by a camera. Example: CCTV frame, IP camera stream, facial recognition input.
13	Lidar	Point cloud or distance measurements produced by a laser-based sensor. Example: 3D point cloud from a rotating lidar, perimeter scanning lidar.
14	Light	Ambient light level or flicker detected by a photodetector. Example: lux measurement, room lighting state, strobe or flicker detection.
15	Log	Structured or unstructured event records produced by systems, applications or security devices. Example: syslog messages, Windows EventLog entries, journald records, firewall logs, application traces.
16	Magnetic	Magnetic field strength or anomaly detected by a magnetometer. Example: door open/close magnetic contact, vehicle detection, magnetic anomaly near equipment.
17	Particles	Concentration of dust, smoke or airborne contaminants measured by a particle sensor. Example: smoke detector, PM2.5 sensor, dust monitoring in a facility.
18	PCAP	Packet capture files stored on disk and analyzed offline. Example: .pcap and .pcapng files produced by tcpdump, dumpcap or Wireshark.

Rank	Keyword	Description
19	Rain	Precipitation detected or measured by a rain sensor. Example: rainfall detection on an outdoor site, rain gauge.
20	Seismic	Ground motion or earthquake-related signals captured by a seismograph. Example: earthquake detection, ground vibration near a datacenter.
21	SNMP	Data retrieved via the Simple Network Management Protocol from managed devices. Example: SNMP GET/WALK results, MIB object values, traps and informs.
22	Temperature	Ambient, device or surface temperature measured by a thermometer or thermal probe. Example: server room temperature, device CPU temperature, cold-chain monitoring.
23	Thermal	Infrared image or heat signature data captured by a thermal sensor. Example: thermal camera frame, heat map of a room or equipment.
24	Vibration	Mechanical oscillation or structural movement measured by an accelerometer or vibration sensor. Example: seismic-grade vibration of a server rack, structural health monitoring of a bridge.
25	Water	Presence, flow or level of water detected by a sensor. Example: leak detection under a raised floor, flood sensor, water flow meter.
26	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 12: enum_analyzer_data

C.5. Analyzer Method

Rank	Keyword	Description
0	Undetermined	Analyser method is undetermed
1	AI	An analyzer that uses machine learning, deep learning, or other

Rank	Keyword	Description
		artificial intelligence techniques to learn normal behavior and detect sophisticated or novel threats.
2	Anomaly	An analyzer that identifies deviations from established norms or baselines without relying on predefined signatures, often flagging unusual patterns in traffic, behavior, or system activity.
3	Behavioral	An analyzer that monitors and analyzes the actions of users, entities, or processes over time to detect malicious or suspicious activities that deviate from expected behavior patterns.
4	Biometric	An analyzer that uses electronic devices to capture and measure unique physical or behavioral characteristics (e.g., fingerprint, iris, voice) for identification or authentication purposes.
5	Blackhole	A method that analyses traffic destined for a non-existent or sinkhole route to identify malicious activity, such as connections to known command-and-control servers or scanning from infected hosts.
6	Contextual	An analyzer that enriches raw events with additional context (e.g., asset value, user role, time of day, business criticality) to prioritize alerts and assess true impact.
7	Correlation	An analyzer that aggregates and examines multiple, disparate data streams or events to identify complex relationships, patterns, sequences, or dependencies that indicate a security incident.
8	Ensemble	An analyzer that combines multiple detection methods (e.g., signature, anomaly, behavioral) to improve accuracy, reduce false positives, and detect threats that single methods might miss.
9	Fingerprinting	An analyzer that creates unique identifiers or "fingerprints"

Rank	Keyword	Description
		for devices, applications, or network stacks to detect spoofing, unauthorized devices, or configuration changes.
10	Frequency	An analyzer that detects incidents based on the rate or regularity of events (e.g., repeated failed logins, rapid-fire requests) exceeding or falling below expected frequencies.
11	Fusion	An analyzer that combines data from multiple heterogeneous sensors and sources to create a comprehensive, high-confidence view of an incident, reducing ambiguity and false alerts.
12	Geolocation	An analyzer that determines the physical location of an asset or event (e.g., login attempt, IP address, device) and flags activities occurring from unexpected or high-risk locations.
13	Graph-based	An analyzer that models relationships between entities (users, devices, processes) as graphs and detects anomalies or attack paths by analyzing connections, dependencies, and traversals.
14	Heat	An analyzer (sensor or device) that detects, measures, and monitors thermal energy (infrared radiation) to identify anomalies like fires, overheating equipment, or human presence.
15	Heuristic	An analyzer that detects potentially unknown threats by using algorithmic logic, rules of thumb, or suspicious characteristics rather than relying on specific signature matches.
16	Honeypot	A decoy system or resource designed to lure, detect, and analyze malicious activity by mimicking a legitimate target, diverting attackers away from real assets.
17	Hygiene	An analyzer that continuously checks systems and

Rank	Keyword	Description
		configurations against security best practices, compliance standards, or hardening guidelines to identify weaknesses or drift.
18	Integrity	An analyzer that monitors critical system components (files, configurations, registry keys) for unauthorized changes, verifying their integrity against a known good baseline.
19	Metadata	An analyzer that examines the data about data (e.g., file creation timestamps, email headers, connection logs) to uncover hidden relationships or suspicious attributes.
20	Monitor	An analyzer that continuously observes a system, network, or environment to track its state, health, or activity, often providing real-time alerts on specific conditions.
21	Movement	An analyzer (sensor or system) that detects, tracks, and quantifies physical motion using technologies like radar, lidar, or video analytics.
22	Orchestration	An analyzer that coordinates and triggers automated response actions based on detected incidents, often integrated with SOAR (Security Orchestration, Automation, and Response) platforms.
23	Pattern	An analyzer that identifies specific sequences, combinations, or recurring arrangements of events or data that indicate malicious activity, even if individual elements appear benign.
24	Policy	An analyzer that evaluates events, configurations, or behaviors against a set of predefined rules, configurations, or compliance requirements to detect violations or misconfigurations.
25	Predictive	An analyzer that uses historical data and modeling to forecast potential future incidents,

Rank	Keyword	Description
		vulnerabilities, or attack vectors before they occur.
26	Protocol	An analyzer that validates network traffic or communications against expected protocol specifications, RFC compliance, or standard behavior to detect anomalies or malicious variations.
27	Recon	An analyzer that actively or passively probes or monitors an environment to discover assets, services, or vulnerabilities, often as part of a defensive assessment or adversary simulation.
28	Reputation	An analyzer that evaluates the trustworthiness of an entity (e.g., IP address, domain, file hash) by checking it against known threat intelligence lists, blocklists, or reputation scores.
29	Rule-based	An analyzer that applies conditional logic (if-then-else statements) defined by experts to correlate events and generate alerts based on specific combinations of conditions.
30	Sequence	An analyzer that detects threats by examining the order and timing of events, identifying attack chains or kill chain progressions (e.g., scan # exploit # installation # C2).
31	Signature	An analyzer that detects known threats by matching events or patterns against a database of specific signatures, hashes, or Indicators of Compromise (IoCs).
32	Statistical	An analyzer that detects anomalies by establishing a baseline of normal behavior and identifying events that deviate significantly from expected statistical parameters.
33	Tarpit	A mechanism that intentionally slows down or delays suspicious connections (e.g., network connections or login

Rank	Keyword	Description
		attempts) to hinder automated attacks and scanning.
34	Threat Intelligence	An analyzer that ingests and matches internal events against external threat feeds, IoC lists, and adversary TTPs (Tactics, Techniques, and Procedures) to identify known threats.
35	Threshold	An analyzer that detects incidents by comparing a metric or count (e.g., number of failed logins, traffic volume) against a predefined limit or threshold.
36	Trend	An analyzer that monitors data over extended periods to identify gradual changes, emerging patterns, or long-term shifts that may indicate evolving threats or security degradation.
37	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 13: Analyzer Methods

C.6. Source Category

Rank	Keyword	Description
0	Undetermined	Source Category is undetermined.
1	Acoustic.Sound.GlassBreak	Specific frequencies (glass breaking, gunshots)
2	Acoustic.Sound.Infrasound	Infrasound (nuclear explosions, volcanic activity)
3	Acoustic.Sound.PressurizedLeak	Leaks in pressurized pipes (high-frequency hiss)
4	Acoustic.Sound.SonarEcho	Sonar echoes (submarines, fish, seafloor mapping)
5	Acoustic.Sound.UltrasonicNoise	Ultrasonic noise (for pest detection or leak detection)
6	Acoustic.Vibration.Footstep	Footsteps (geophones or seismic sensors)
7	Acoustic.Vibration.Seismic	Seismic activity (earthquakes, tremors)
8	Acoustic.Vibration.StructuralCrack	Structural integrity (cracks via acoustic emission)
9	Acoustic.Vibration.VehicleVibe	Vehicle movement (magnetic or seismic)
10	Acoustic.Vibration.VibAnalysis	Machinery imbalance (vibration analysis)

Rank	Keyword	Description
11	ChemBio.Gases.CO	Carbon Monoxide (CO)
12	ChemBio.Gases.CWA	CWA (Chemical Warfare Agents - Nerve gas, Blister agents)
13	ChemBio.Gases.Freon	Refrigerant leaks (Freon)
14	ChemBio.Gases.Hydrogen	Hydrogen
15	ChemBio.Gases.Methane	Natural Gas / Methane
16	ChemBio.Gases.O2Level	Oxygen levels (deficiency or enrichment)
17	ChemBio.Gases.Propane	Propane / LPG
18	ChemBio.Gases.Radon	Radon
19	ChemBio.Gases.VOC	Volatile Organic Compounds (VOCs) - paint fumes, off-gassing
20	ChemBio.Properties.ORP	Oxidation-Reduction Potential (ORP)
21	ChemBio.Properties.pHLevel	pH levels (acidity/alkalinity)
22	ChemBio.Properties.Salinity	Salinity
23	ChemBio.Threats.Anthrax	Anthrax spores
24	ChemBio.Threats.Botulinum	Botulinum toxin
25	ChemBio.Threats.Pathogen	Pathogens in water (E. coli, Cholera)
26	ChemBio.Threats.Ricin	Ricin
27	Cyber.Application.SQLQuery	Malicious code injected into a database query (SQL Injection).
28	Cyber.Application.XSSPayload	The script injected into a webpage to steal data or redirect users.
29	Cyber.Cloud.APIToken	A compromised key used to access cloud resources (AWS, Azure) or SaaS platforms.
30	Cyber.Cloud.BucketName	The specific cloud storage location where data was stored or exfiltrated from.
31	Cyber.Cloud.CloudResourceID	The unique ID of the compromised virtual machine or database.
32	Cyber.Cloud.OAuthApp	A malicious third-party app granted permissions to a user's data.
33	Cyber.Email.EmailAuthResult	The authentication result showing if the email passed security checks.
34	Cyber.Email.MessageID	A unique identifier for the email used to track it across mail servers.
35	Cyber.Email.RecipientEmail	The target email address of the attack.
36	Cyber.Email.ReplyToEmail	The address where replies go; often different from the sender for tracking.

Rank	Keyword	Description
37	Cyber.Email.ReturnPath	The address where non-delivery receipts go; used in spoofing analysis.
38	Cyber.Email.SenderEmail	The "From" address used in a phishing campaign.
39	Cyber.Email.XOrigIP	The original IP address of the sender (if recorded by the mail server).
40	Cyber.Endpoint.CmdLine	The exact command line used to execute the malware; often reveals attacker intent.
41	Cyber.Endpoint.DLL	The name of a malicious library loaded into a legitimate process (DLL Sideloading).
42	Cyber.Endpoint.FileHash	The cryptographic fingerprint of the malicious file (SHA256, MD5).
43	Cyber.Endpoint.FilePath	The location on the file system where the malware resides.
44	Cyber.Endpoint.MemoryArtifact	Code or processes that exist only in RAM (volatile memory) and not on disk.
45	Cyber.Endpoint.PID	The unique identifier of the malicious process at the time of analysis.
46	Cyber.Endpoint.ProcessName	The name of the malicious executable file running on the endpoint.
47	Cyber.Endpoint.RegistryKey	The Windows Registry path used for persistence or configuration changes.
48	Cyber.Endpoint.ScheduledTask	The name of a task created by the attacker for persistence or execution.
49	Cyber.Endpoint.ServiceName	The name of a malicious service installed to run the malware.
50	Cyber.Forensics.NetFlow	Metadata about the connection (who talked to whom, when, and data volume).
51	Cyber.Forensics.PacketPayload	The raw data being transmitted containing passwords or exfiltrated data.
52	Cyber.Hardware.DeviceSerial	The unique identifier of a specific compromised laptop or mobile device.
53	Cyber.Hardware.USBDeviceID	The specific make/model/ serial of a USB drive used to introduce malware.
54	Cyber.Identity.LoginTimestamp	The specific time of an unusual login (e.g., 3:00 AM from a foreign location).

Rank	Keyword	Description
55	Cyber.Identity.Username	The specific user account that was compromised or used for lateral movement.
56	Cyber.Network.IPv4	The source IP of an attacker, a Command & Control (C2) server, or a malicious scan.
57	Cyber.Network.IPv6	The source IP in IPv6 format; increasingly used to bypass IPv4-based allowlists.
58	Cyber.Network.MAC	The physical device identifier on a local network; useful if the attacker is on-site.
59	Cyber.Network.Port	The specific source or destination port used (e.g., 445 for ransomware spreading).
60	Cyber.Network.Protocol	The network protocol used; anomalous use may indicate tunneling (e.g., DNS tunneling).
61	Cyber.Web.Domain	The domain hosting malware or Command & Control infrastructure (e.g., evil.com).
62	Cyber.Web.HTTPMethod	Unusual HTTP methods like PUT or DELETE used to manipulate data.
63	Cyber.Web.ReferrerHeader	The page the user was on before clicking the malicious link.
64	Cyber.Web.SessionID	A hijacked session token used to impersonate a user without a password.
65	Cyber.Web.URIPath	The specific directory or file requested on the web server.
66	Cyber.Web.URL	The full malicious link pointing to a payload or phishing page.
67	Cyber.Web.UserAgent	The browser or application signature used by the attacker when connecting.
68	ElectroMag.Infrared.GasLeakIR	Gas leaks (specific gases absorb specific IR wavelengths)
69	ElectroMag.Infrared.HeatTemp	Heat (temperature of objects)
70	ElectroMag.Infrared.MoistureContent	Moisture content (in materials)
71	ElectroMag.Infrared.ThermalGradient	Thermal gradients (heat loss in buildings)
72	ElectroMag.Infrared.WarmBody	Warm bodies (humans, animals) via Passive Infrared (PIR)
73	ElectroMag.Ultraviolet.BioContamUV	Biological contamination
74	ElectroMag.Ultraviolet.CoronaArc	Corona discharge (electrical arcing)
75	ElectroMag.Ultraviolet.FlameUV	Flames (fire detection)
76	ElectroMag.Ultraviolet.ForgeryUV	Forgery (detecting UV security features on documents)

Rank	Keyword	Description
77	ElectroMag.Light.BiometricEye	Biological signatures (retinas, irises) - Failed recognition indicates intrusion attempt
78	ElectroMag.Light.EdgesShapes	Edges and shapes (for machine vision)
79	ElectroMag.Light.Flicker	Flicker (from faulty lighting or machinery)
80	ElectroMag.Light.LightColor	Color and wavelength (spectral content)
81	ElectroMag.Light.LightPresence	Presence of light (ambient brightness)
82	ElectroMag.Light.MotionOptical	Motion (optical flow, changes in pixel arrays)
83	ElectroMag.Light.ObjectsGeneric	Specific objects (via computer vision: faces, vehicles, text, barcodes)
84	Force.Atmospheric.Altitude	Altitude (in aircraft)
85	Force.Atmospheric.BaroPress	Barometric pressure (weather changes)
86	Force.Atmospheric.VacuumLevel	Vacuum levels (in manufacturing)
87	Force.Fluid.Airflow	Airflow (differential pressure)
88	Force.Fluid.WaterPressure	Water pressure (for leak detection via pressure drop)
89	Force.Touch.GripStrength	Grip strength
90	Force.Touch.PhysicalContact	Physical contact (bumpers on robots, touch screens)
91	Force.Touch.TirePressure	Tire pressure
92	Force.Touch.Weight	Weight (scales, load cells)
93	Radiation.General.Alpha	Alpha particles
94	Radiation.General.Beta	Beta particles
95	Radiation.General.Cosmic	Cosmic rays
96	Radiation.General.Gamma	Gamma rays
97	Radiation.General.Neutron	Neutron radiation
98	Radiation.General.RadonDecay	Radon gas (decay products)
99	Radiation.General.Xray	X-rays
100	Magnetic.General.CurrentFlow	Current flow (Hall effect sensors, clamp meters)
101	Magnetic.General.ESD	Electrostatic discharge
102	Magnetic.General.Ferrous	Ferrous metals (via magnetic disturbance)
103	Magnetic.General.MagneticAnomaly	Magnetic anomalies (submarines, buried structures)
104	Magnetic.General.Orientation	Orientation (Compass / Magnetometer - detecting Earth's magnetic field)
105	Magnetic.General.Powerline	Power lines (AC magnetic fields)
106	Magnetic.General.Voltage	Voltage (non-contact voltage testers)
107	Object.Indoor.Container	Bottles / Cups / Cutlery
108	Object.Indoor.Document	Books / Papers / Documents

Rank	Keyword	Description
109	Object.Indoor.Electronics	Electronics (laptops / phones / TVs)
110	Object.Indoor.Furniture	Furniture (chairs / tables / sofas)
111	Object.Indoor.Weapon	Weapons (guns / knives)
112	Object.LivingBeings.Animal	Animals (general wildlife/ livestock/pets)
113	Object.LivingBeings.BodyPose	Pose (skeleton keypoints / body position)
114	Object.LivingBeings.EyeGaze	Eyes (gaze tracking / eye blink)
115	Object.LivingBeings.Face	Faces (facial recognition)
116	Object.LivingBeings.HandGesture	Hands (hand gestures / raising hand)
117	Object.LivingBeings.Human	Humans (person/people detection)
118	Object.Outdoor.Building	Buildings / Houses
119	Object.Outdoor.Pothole	Potholes / Road cracks
120	Object.Outdoor.RoadSign	Road signs / Traffic lights
121	Object.Outdoor.Vegetation	Trees / Plants / Vegetation
122	Object.SpecificScenes.FireVisual	Fire / Flames (visual spectrum)
123	Object.SpecificScenes.Flood	Flooding / Water accumulation
124	Object.SpecificScenes.SmokeVisual	Smoke plume (visual spectrum)
125	Object.SpecificScenes.Snow	Snow coverage
126	Object.Vehicles.Aircraft	Aircraft (drones / planes / helicopters)
127	Object.Vehicles.Boat	Boats / Ships
128	Object.Vehicles.Bus	Buses
129	Object.Vehicles.Car	Cars / Automobiles
130	Object.Vehicles.LicensePlate	License plates (Automatic Number Plate Recognition - ANPR)
131	Object.Vehicles.Moto	Motorcycles / Bicycles
132	Object.Vehicles.Truck	Trucks
133	Particulate.Aerosols.Bioaerosol	Viral or bacterial particles (bio-aerosol detectors)
134	Particulate.Aerosols.Dust	Airborne dust (particulate counters)
135	Particulate.Aerosols.Pollen	Pollen and allergens
136	Particulate.Aerosols.SmokeIonization	Combustion particles (Ionization smoke detectors)
137	Particulate.Aerosols.SmokePhotoelectric	Visible smoke obscuration (Photoelectric smoke detectors)
138	Particulate.Liquids.ChemicalSpill	Chemical spills (specific conductivity or pH changes)
139	Particulate.Liquids.FuelSheen	Fuel/Oil sheen on water
140	Particulate.Liquids.LiquidLevel	Level of liquid in a tank (ultrasonic, capacitive, float)
141	Particulate.Liquids.WaterLeak	Water leakage (conductivity or float switches)
142	Particulate.Solids.DensityChange	Density changes (in manufacturing)
143	Particulate.Solids.ForeignObject	Foreign objects in food (X-ray machines)

Rank	Keyword	Description
144	Particulate.Solids.GrainFlow	Grain flow (in agriculture)
145	Particulate.Solids.Landmine	Landmines/UXO (Ground Penetrating Radar)
146	Particulate.Solids.Metal	Metal (Metal detectors via eddy currents)
147	Software.Detected.Crowd	Crowd formation
148	Software.Detected.CyberAttack	Cyber-attack (Anomalous network traffic patterns)
149	Software.Detected.Drowsiness	Drowsiness (Eye closure detection via camera)
150	Software.Detected.Explosion	Explosion (Sudden light flash + pressure wave + sound)
151	Software.Detected.FailedBiometric	Failed retina/iris scan (multiple failures may indicate intrusion)
152	Software.Detected.FireFusion	Fire (Combination of smoke, heat, and flicker)
153	Software.Detected.Intrusion	Intrusion (Crossing a virtual line)
154	Software.Detected.Loitering	Loitering (Person staying too long)
155	Software.Detected.Tailgating	Tailgating (Following someone through a door)
156	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 14: Source Categories

C.7. Target Category

Rank	Keyword	Description
0	Undetermined	Target Category is undetermined
1	Commercial.ClientBase	Client portfolio, active contracts
2	Commercial.Partnerships	Strategic alliances, partner networks
3	Commercial.Reputation	Brand image, online reputation, goodwill
4	Financial.Cash	Liquidities, bank accounts
5	Financial.Investments	Shares, bonds, holdings in subsidiaries
6	Financial.Receivables	Customer receivables, promissory notes
7	Furniture.Fittings	Lighting, partitions, flooring
8	Furniture.HVAC	Air conditioning, heating systems (non-industrial)
9	Furniture.OfficeFurniture	Desks, chairs, filing cabinets
10	Human.EmployerBrand	Employment contracts, recruitment appeal

Rank	Keyword	Description
11	Human.Organization	Work processes, internal know-how
12	Human.Skills	Employee expertise, team competencies
13	Infrastructure.CivilWorks	Bridges, loading docks, quays
14	Infrastructure.Foundations	Special foundations, internal roads, pavements
15	Infrastructure.Networks	Buried water, gas, electricity, fiber networks
16	IP.Copyrights	Software code, manuals, artistic works
17	IP.Patents	Registered patents, inventions
18	IP.Trademarks	Brand names, logos, registered designs
19	IT.Endpoints	Computers, printers, scanners
20	IT.Hardware	Servers, storage arrays, cabling
21	IT.Telecom	WiFi access points, phones, smartphones
22	Logistics.Containers	Containers, trailers, wagons
23	Logistics.SpecialVehicles	Mobile cranes, aerial work platforms
24	Logistics.Vehicles	Cars, trucks, forklifts
25	Marketing.Content	Market studies, ad campaigns, social media
26	Marketing.Domains	Website URLs, domain names
27	Marketing.Identity	Visual identity, logo, brand guidelines
28	Production.HeavyEquipment	Presses, furnaces, compressors, generators
29	Production.Machinery	Assembly lines, industrial robots, conveyors
30	Production.MachineTools	Lathes, milling machines, industrial 3D printers
31	RealEstate.Buildings	Factories, warehouses, offices, commercial premises
32	RealEstate.Land	Unbuilt land, forests, or agricultural grounds
33	RealEstate.SpecialStructures	Hangars, silos, parking lots
34	Software.Applications	ERP, CRM, SaaS platforms
35	Software.Data	Customer databases, supplier lists, prospects
36	Software.Licenses	Software usage rights, subscriptions
37	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 15: Target Categories

Appendix D. Examples

This section contains several examples of events/incidents which may be described using the IDMEF Data Model defined in.

For each example, the serialization method listed in Section 5 was used on the original IDMEF message to produce a JSON representation.

D.1. Physical intrusion

Listing 1 describes an incident where an unidentified man was detected on company premises near the building where server room A is located.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b1",
  "Description": "Potential intruder detected",
  "Type": "Physical",
  "Priority": "Low",
  "Status": "Incident",
  "Cause": "Malicious",
  "CreateTime": "2021-05-10T16:52:13.075994+00:00",
  "StartTime": "2021-05-10T16:52:13+00:00",
  "Category": [
    "Access.Forced"
  ],
  "Analyzer": {
    "Name": "BigBrother",
    "Hostname": "bb.acme.com",
    "Model": "Big Brother v42",
    "Category": [
      "PHY.HAR",
      "FRC.FR-CAM"
    ],
    "Data": [
      "Images"
    ],
    "Method": [
      "Movement",
      "Biometric",
      "AI"
    ],
    "IP": "192.0.2.1"
  },
  "Sensor": [
    {
      "IP": "192.0.2.2",
      "Name": "Camera #23",
      "Model": "SuperDuper Camera v1",
      "Location": "Hallway to server room A1"
    }
  ],
  "Source": [
    {
      "Note": "Black Organization, aka. APT 4869"
    }
  ],
  "Attachment": [
    {
      "Name": "wanted",
      "FileName": "fbi-wanted-poster.jpg",
      "Size": 1234567,
      "Ref": ["https://www.fbi.gov/wanted/topten"],
      "ContentType": "image/jpg",
      "ContentEncoding": "base64",
      "Content": "..."
    },
    {
      "Name": "pic01",
      "Note": "Hi-res picture showing John Doe near server room A1",
      "ExternalURI": ["ftps://192.0.2.1/cam23/20210510165211.jpg"],
      "ContentType": "image/jpg"
    }
```

D.2. Cyberattack

Listing 2 describes an incident related to a potential bruteforce attack against the "root" user account of the server at 192.0.2.2 and 2001:db8::/32.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b2",
  "Description": "Potential bruteforce attack on root user account",
  "Type": "Cyber",
  "Priority": "Medium",
  "CreateTime": "2021-05-10T16:55:29.196408+00:00",
  "StartTime": "2021-05-10T16:55:29+00:00",
  "Category": [
    "Access.Forced"
  ],
  "Analyzer": {
    "Name": "SIEM",
    "Hostname": "siem.acme.com",
    "Model": "Concerto SIEM 5.2",
    "Category": [
      "SIEM.SIEM",
      "SIEM.LOG"
    ],
    "Data": [
      "Log"
    ],
    "Method": [
      "Monitor",
      "Signature"
    ],
    "IP": "192.0.2.1"
  },
  "Sensor": [
    {
      "IP": "192.0.2.5",
      "Name": "syslog",
      "Hostname": "www.acme.com",
      "Model": "rsyslog 8.2110",
      "Location": "Server room A1, rack 10"
    }
  ],
  "Target": [
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "Location": "Server room A1, rack 10",
      "User": "root"
    },
    {
      "IP": "2001:db8::/32",
      "Hostname": "www.acme.com",
      "Location": "Server room A1, rack 10",
      "User": "root"
    }
  ]
}
```

D.3. Server outage

Listing 3 describes an incident where the webserver at "www.example.com" encountered some kind of failure condition resulting in an outage.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b3",
  "Description": "A server did not reply to an ICMP ping request",
  "Type": "Availability",
  "Priority": "Medium",
  "Status": "Incident",
  "Cause": "Unknown",
  "CreateTime": "2021-05-10T16:59:11.875209+00:00",
  "StartTime": "2021-05-10T16:59:11.875209+00:00",
  "Category": [
    "Availability.Outage"
  ],
  "Analyzer": {
    "Name": "NMS",
    "Hostname": "nms.example.com",
    "Model": "Concerto NMS 5.2",
    "Category": [
      "SIEM.NMS"
    ],
    "Data": [
      "Network"
    ],
    "Method": [
      "Monitor"
    ],
    "IP": "192.0.2.1"
  },
  "Target": [
    {
      "IP": "192.168.1.2",
      "Hostname": "www.acme.com",
      "Service": "website",
      "Location": "Server room A1, rack 10"
    }
  ]
}
```

D.4. Combined incident

Listing 4 describes a combined incident resulting from the correlation of the previous physical, cyber and availability incidents.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b4",
  "Description": "Intrusion and Sabotage detected",
  "Type": "Combined",
  "Priority": "High",
  "Status": "Incident",
  "Cause": "Malicious",
  "CreateTime": "2021-05-10T16:59:15.075994+00:00",
  "StartTime": "2021-05-10T16:52:11+00:00",
  "Category": [
    "Access.Unauthorized",
    "Sabotage.Data",
  ,
  "CorrelID": [
    "819df7bc-35ef-40d8-bbee-1901117370b1",
    "819df7bc-35ef-40d8-bbee-1901117370b2",
    "819df7bc-35ef-40d8-bbee-1901117370b3"
  ],
  "Analyzer": {
    "Name": "Correlator",
    "Hostname": "correlator.acme.com",
    "Model": "Concerto Hybrid Correlator v5.2",
    "Category": [
    ],
    "Data": [
      "Alert"
    ],
    "Method": [
      "Correlation"
    ],
    "IP": "192.0.2.1"
  },
  "Source": [
    {
      "Note": "Black Organization, aka. APT 4869"
    }
  ],
  "Target": [
    {
      "Location": "Server room A1"
    },
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "User": "root"
    },
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "Service": "website"
    }
  ]
}
```

D.5. Hazard incident

Listing 5 describes a heavy snow storm announced in 48h on Paris and Versailles.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b1",
  "Description": "Snow storm forecast",
  "Priority": "Low",
  "Status": "Incident",
  "Cause": "Hazard",
  "Confidence": 0.8,
  "CreateTime": "2021-05-10T16:52:13.075994+00:00",
  "StartTime": "2021-05-12T10:00:00+00:00",
  "EndTime": "2021-05-12T10:00:00+00:00",
  "Type": ["Physical"],
  "Category": [
    "Meteo.Snow",
    "Meteo.Wind",
    "Meteo.Cold"
  ],
  "Analyzer": {
    "Name": "Weather Monitor",
    "Hostname": "weather.acme.com",
    "IP": "192.0.2.1"
  },
  "Source": [
    {
      "Note": "Heavy snow storm coming from North"
    }
  ],
  "Target": [
    {
      "GeoLocation": "48.8584,2.2945",
      "UnLocation": "FR PAR",
      "Location": "Acme Paris Site"
    },
    {
      "GeoLocation": "48.8019,2.1301",
      "UnLocation": "FR VER",
      "Location": "Acme Versailles Site"
    }
  ]
}
```

Appendix E. JSON Validation Schema (Non-normative)

Listing 5 contains a JSON Schema that can be used to validate incoming IDMEF messages prior to processing. Please note that extraneous linebreaks have been included due to formatting constraints.

```
{
    "description": "JSON schema for the Intrusion Detection Message
 Exchange Format (IDMEF) version 2 (revision 2.D.V10-dev)",
    "properties": {
        "Version": {
            "description": "The version of the IDMEF format in use by this
 alert. During the drafts tuning period the version is equal to the draft
 version. Therefore it is \"2.D.V0X\" for Draft V0X.",
            "enum": [
                "2.D.V10-dev"
            ]
        },
        "ID": {
            "description": "Unique identifier for the alert.",
            "$ref": "#/definitions/uuidType"
        },
        "OrganisationName": {
            "description": "Corporate/Main Office Organisation Name Useful
 if alerts are sent to a multi-organisation central incident detection
 manager. Example: ACME Corporation",
            "type": "string"
        },
        "OrganisationId": {
            "description": "Corporate/Main Office Organisation ID. Where
 possible official organisation ID manage by national authority. Useful
 if alerts are sent to a multi-organisation central incident detection
 manager. This ID has to be chosen depending on the overall detection
 perimeter and the nature of the monitored organisation (Private/Public,
 Commercial, International, etc.) Examples: OrganisationId in France could
 be SIREN, in England could be CR, Germany could be Handelsregisternummer,
 Spain could be CIF, Italia could be Partita IVA, USA could be EIN, etc.
 Commercial OrganisationId in Europe could be V.A.T ID",
            "type": "string"
        },
        "EntityName": {
            "description": "Entity Name, monitored by the organisation,
 where the incident occurred. Could be a town, region or country name
 or an internal name. Could also be the name of a customer for a MSSP
 centralizing it's customer incidents in a single system. Do not repeat
 the organisation name in the EntityName Example: - ACME HeadQuaters
 is located in Paris France and has a local office in India - If the
 incident occurred in the local office: \"OrganisationName\": \"ACME\" ,
 \"EntityName\": \"India\" - If the incident occurred in the headquaters:
 \"OrganisationName\": \"ACME\", \"EntityName\": \"Headquaters\" (or
 \"Paris\")",
            "type": "string"
        },
        "EntityId": {
            "description": "Entity ID, monitored by the organisation,
 where the incident occurred. Useful if organisation and entity are not
 directly linked, like a customer and a MSSP.",
            "type": "string"
        },
        "EntitySector": {
            "description": "The economic sector(s) and sub-sector(s)
 in which the entity operates. Values follow the dot notation
 sector.subsector based on the critical infrastructure taxonomy defined
 in the NIS2 Directive and CER (Critical Entities Resilience) Directive.
 This attribute enables sector-based correlation, regulatory compliance
 reporting, and risk context for incident detection.",
```

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